Theory of Generalized Linear Model Notes

Yunhe Liu

Contents

good luck :)

1 Bootstrap

The Bootstrap Principle

Suppose $\mathcal{D} \sim G$, $\psi = \psi(G)$ is some parameter of G, and $T = T(\mathcal{D})$ is a statistic aimed at estimating ψ. The sampling distribution of T is typically unknown because G is unknown. The bootstrap gets around this problem by:

1. Estimating G with some $\hat{G}$ based on the observed data; and

2. Using the sampling distribution of $T^\star = T(\mathcal{D}^\star)$, where $\mathcal{D}^\star \sim \hat{G}$, as an estimate of the sampling distribution of T.

Remark. The logic of the bootstrap can be summarized in a chain:

$$\mathcal{D} \sim G \quad \xrightarrow{\text{estimate}} \quad G \approx \hat{G} \quad \xrightarrow{\text{simulate}} \quad \mathcal{D}^\star \sim \hat{G} \quad \xrightarrow{\text{compute}} \quad T^\star = T(\mathcal{D}^\star).$$

We then use the distribution of $T^\star$ to approximate the sampling distribution of $T(\mathcal{D})$. The most important point here is $G \approx \hat{G}$. It means we don't know the data-generating process G, so we approximate G through $\hat{G}$ and regard $\hat{G}$ as our data-generating process to do downstream analysis.
The Nonparametric Bootstrap. Suppose $\mathcal{D} = (X_1, \ldots, X_N)$ with $X_i \overset{\text{iid}}{\sim} F$, so that $G = F^N$. The nonparametric bootstrap takes $\hat{F} = F_N = N^{-1} \sum_{i=1}^N \delta_{X_i}$ (the empirical distribution), so that $\hat{G} = \hat{F}^N$. In practice, sampling from $\hat{F}$ means sampling N observations *with replacement* from the original data.
The Parametric Bootstrap. If we are willing to assume a parametric model $\{G_\theta : \theta \in \Theta\}$, then the parametric bootstrap takes $\hat{G} = G_{\hat{\theta}}$ where $\hat{\theta}$ is typically the MLE. For example, if $X_i \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$, then we would bootstrap from $\text{Normal}(\bar{X}, s^2)$.
Computing the Sampling Distribution via Monte Carlo. In all but the simplest settings, the sampling distribution of $T^\star$ is approximated by simulation: (1) sample many pseudo-datasets $\mathcal{D}_1^\star, \ldots, \mathcal{D}_B^\star$ independently from $\hat{G}$; (2) compute $T_b^\star = T(\mathcal{D}_b^\star)$ for each; (3) approximate the distribution of $T^\star$ with the empirical distribution $B^{-1} \sum_{b=1}^B \delta_{T_b^\star}$.

Bootstrapping Functionals

Suppose $X_1, \ldots, X_N \overset{\text{iid}}{\sim} F$. Many estimands of interest can be written as *functionals* of F, denoted $\psi(F)$. The plug-in principle gives a natural estimator $\hat{\psi} = \psi(\hat{F})$: if $\hat{F}$ is close to F, then $\psi(\hat{F})$ should be close to $\psi(F)$.

Bootstrapping Linear Functionals (Exercise 1). Suppose that $X_1, \ldots, X_N \overset{\text{iid}}{\sim} F$ and let $\psi(F)$ denote the population mean of F, i.e., $\psi(F) = \mathbb{E}_F\big[X_i\big] = \int x \, F(dx)$. We consider bootstrapping the sample mean $\bar{X}_N = N^{-1} \sum_{i=1}^N X_i$ using the approximation $\hat{F} = F_N$. That is, we consider the sampling distribution of $\bar{X}^\star = N^{-1} \sum_{i=1}^N X_i^\star$ where $X_1^\star, \ldots, X_N^\star$ are sampled independently from F_N. **Note: none of the answers to these questions involve considering simulated datasets.**
(a) What is $\psi(F_N)$?
Answer. $\psi(F_N) = \int x \, F_N(dx) = \bar{X}_N$.
(b) The *actual* bias of $\bar{X}_N$ is $\mathbb{E}_F\big[\bar{X}_N - \psi(F)\big] = 0$. What is the *bootstrap estimate* of the bias $\mathbb{E}_{F_N}\big[\bar{X}_N^\star - \bar{X}\big]$?
Answer. $\mathbb{E}_{F_N}\big[\bar{X}_N^\star - \bar{X}\big] = \bar{X} - \bar{X} = 0$. So the bootstrap correctly identifies the sample mean as unbiased (for itself).

(c) The variance of $\bar{X}_N$ is σ_F^2/N where σ_F^2 is $\mathrm{Var}_F(X_i)$. What is the *bootstrap estimate* of the variance of $\bar{X}$, $\mathrm{Var}_{F_N}(\bar{X}^\star)$?
Answer. $\mathrm{Var}_{F_N}(\bar{X}^\star) = \sigma_{F_N}^2/N$, where $\sigma_{F_N}^2 = N^{-1}\sum_i (X_i - \bar{X})^2$ is the variance of the empirical distribution.
(d) A parameter ψ is said to be *linear* if it can be written as $\psi(F) = \int t(x)\, F(dx)$ for some choice of $t(x)$. In this case it is natural to estimate ψ using $\bar{T} = N^{-1}\sum_i t(X_i)$. Write down the bootstrap estimate of the bias and variance of $\bar{T}$ in this setting.
Answer. The bootstrap estimate of the bias is 0, and the bootstrap estimate of the variance is σ_T^2/N where $\sigma_T^2 = N^{-1}\sum_i (t(X_i) - \bar{T})^2$.
Remark. Under $\mathbb{E}_{F_N}[\cdot]$, there are two layers to keep straight. In the **real world**, $X_1, \dots, X_N \overset{\text{iid}}{\sim} F$ and $\bar{X}_N$ is random. In the **bootstrap world**, the data $x_1, \dots, x_N$ have already been observed, so $F_N = N^{-1}\sum_{i=1}^N \delta_{x_i}$ is fixed and $\bar{X}_N$ is a constant; only the bootstrap sample $X_1^\star, \dots, X_N^\star \overset{\text{iid}}{\sim} F_N$ is random. This is why $\mathbb{E}_{F_N}[\bar{X}_N^\star - \bar{X}] = \mathbb{E}_{F_N}[\bar{X}_N^\star] - \bar{X}$: the $\bar{X}$ comes out of the expectation as a constant.
Remark. Linear functionals have straightforward standard errors, so the bootstrap is most useful for *nonlinear* functionals (e.g., the median, the correlation coefficient, the skewness). In addition to estimating the standard error, the bootstrap can also estimate the *bias* of $\hat{\theta}$: define $\widehat{\text{bias}} = \mathbb{E}_{\hat{F}}[\theta^\star] - \hat{\theta}$, and use the bias-corrected estimator $\hat{\theta} - \widehat{\text{bias}}$. This is because at the population level the bias is $E_F[\hat{\theta}] - \theta$, and $E[(\hat{\theta} - \text{bias}) - \theta] = 0$. So, we use a bootstrap to estimate F and also bias by $\widehat{\text{bias}} = \mathbb{E}_{\hat{F}}[\theta^\star] - \hat{\theta}$.

Bootstrap Variance Estimation

To approximate the variance (or standard error) of a statistic T:

1. Draw $\mathcal{D}^\star \sim \hat{G}$ (e.g., sample $X_1^\star, \dots, X_N^\star$ with replacement from $\hat{F}$).
2. Compute $T^\star = T(\mathcal{D}^\star)$.
3. Repeat B times to get $T_1^\star, \dots, T_B^\star$.
4. Let $v_{\text{boot}} = \frac{1}{B}\sum_{b=1}^B (T_b^\star - \bar{T}^\star)^2$ where $\bar{T}^\star = B^{-1}\sum_b T_b^\star$. The bootstrap standard error is $\text{se}_{\text{boot}} = \sqrt{v_{\text{boot}}}$.

Types of Bootstrap Confidence Intervals

The Normal Interval

If T is approximately normally distributed and centered at ψ, we can use

$$T \pm z_{\alpha/2}\sqrt{v_{\text{boot}}}.$$

This works well when T is approximately normal, but can be misleading otherwise.

The Basic Percentile Interval

Take $(T_{\alpha/2}^\star, T_{1-\alpha/2}^\star)$ where $T_\gamma^\star$ denotes the 100γth percentile of the bootstrap distribution. This amounts to pretending the bootstrap distribution of $T^\star$ functions like a Bayesian posterior distribution, treating the bootstrapped statistics as if they were random samples of the true ψ. But the true ψ is not random for a Frequentist! Despite feeling intuitive, this interval is surprisingly hard to justify rigorously.

Why Pivots Matter (Exercise 2)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, 1)$ and let $\psi = e^{\mu}$ and $T = e^{\bar{X}_n}$ be the MLE of ψ. This exercise illustrates why bootstrapping a pivotal quantity outperforms bootstrapping the statistic directly.

Three approaches to a confidence interval for ψ.

- **Delta method:** $T \approx \text{Normal}\big(e^{\mu},\, e^{2\mu}/n\big)$, giving the interval $T \pm z_{\alpha/2}\, \hat{s}$ where $\hat{s} = e^{\bar{X}}/\sqrt{n}$.
- **Nonparametric bootstrap:** resample $X_1, \ldots, X_n$ with replacement from $\hat{F} = F_n$, compute $T_b^\star = e^{\bar{X}_b^\star}$, and form the normal interval $T \pm z_{\alpha/2}\sqrt{v_{\text{boot}}}$.
- **Parametric bootstrap:** sample from $\text{Normal}(\bar{X}_n, 1)$ instead of $\hat{F}$, and form the same type of interval.

Comparing bootstrap distributions to the truth. If we plot histograms of $T^\star$ from both the parametric and nonparametric bootstraps alongside the true sampling distribution of T and the delta method normal approximation, we see a clear discrepancy: both bootstrap distributions are centered at $T = e^{\bar{X}}$, while the true sampling distribution of T is centered at $\psi = e^{\mu}$. So the bootstrap distribution is **location-shifted** by $T - \psi$. The delta method normal is also shifted, being centered at T rather than ψ. None of the three approximations align well with the true distribution.

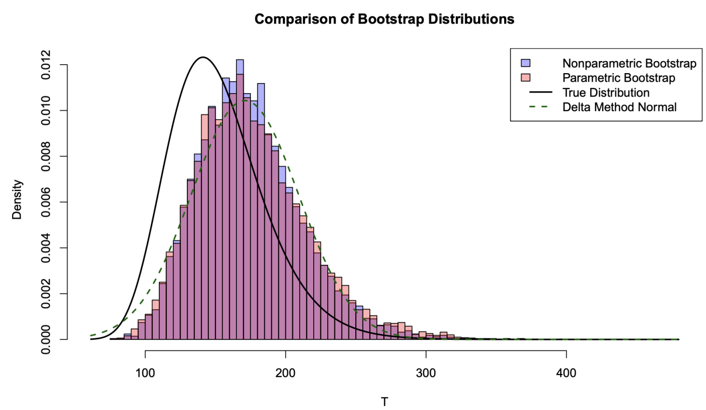

Figure 1: Bootstrapping $T^\star$ directly: both bootstrap distributions and the delta method normal are location-shifted relative to the true sampling distribution of T.

Bootstrapping the centered quantity $T^\star - T$. The location shift suggests that we should not bootstrap $T^\star$ directly, but instead bootstrap the *centered* quantity $T^\star - T$ as an approximation to $T - \psi$. When we compare the bootstrap distribution of $T^\star - T$ to the true distribution of $T - \psi$, the approximation improves dramatically: the location shift is removed, and both the parametric and nonparametric bootstrap distributions now closely track the true distribution. The delta method normal, centered at 0 with variance $e^{2\bar{X}}/n$, also improves but remains somewhat too symmetric compared to the skewed true distribution.

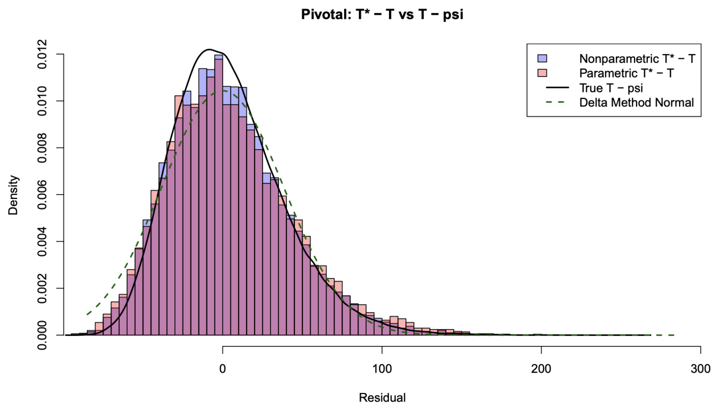

Figure 2: Bootstrapping the centered quantity $T^\star - T$: the location shift is removed and both bootstrap distributions closely approximate the true distribution of $T - \psi$.

From the centered quantity to a confidence interval. Define $\zeta = T - \psi$. Let $\zeta_{\alpha/2}$ and $\zeta_{1-\alpha/2}$ denote its $\alpha/2$ and $1-\alpha/2$ quantiles. Then

$$\Pr\big(\zeta_{\alpha/2} \le T - \psi \le \zeta_{1-\alpha/2}\big) = 1 - \alpha \quad \iff \quad \Pr\big(T - \zeta_{1-\alpha/2} \le \psi \le T - \zeta_{\alpha/2}\big) = 1 - \alpha.$$

In practice, we estimate $\zeta_{\alpha/2}$ and $\zeta_{1-\alpha/2}$ from the bootstrap distribution of $T^\star - T$ (since in the bootstrap world, T plays the role of ψ). The closer $\zeta = T - \psi$ is to being truly pivotal, the better this interval performs. Bootstrapping T directly conflates estimation error with the location of ψ; bootstrapping $T - \psi$ (via its bootstrap analog $T^\star - T$) isolates the estimation error, which is what we actually need for a confidence interval.

A Better Pivot (Exercise 3 & 4)

While we saw an improved approximation for $T - \psi$, argue that this is nevertheless not a pivotal quantity. Propose a pivotal quantity $S(T, \psi)$ which is more suitable for bootstrapping.

Why $T - \psi$ is not pivotal. Recall that a quantity $S(T, \psi)$ is *pivotal* if its sampling distribution does **not** depend on the unknown parameter(s) (here ψ or μ). Consider

$$T - \psi = e^{\bar{X}} - e^{\mu}.$$

We can rewrite this as

$$T - \psi = e^{\mu}\big(e^{\bar{X}-\mu} - 1\big) = \psi\big(e^{\bar{X}-\mu} - 1\big).$$

Since $\bar{X} - \mu \sim \text{Normal}(0, 1/n)$, the random term $\big(e^{\bar{X}-\mu} - 1\big)$ has a distribution that does not depend on μ. However, $T - \psi$ multiplies this term by $\psi = e^{\mu}$, so the *scale* of the distribution depends on ψ. Therefore $T - \psi$ is not pivotal.

Constructing an exact pivot via the log scale. A more natural approach is to work on the log scale. Note that

$$\log T = \log(e^{\bar{X}}) = \bar{X}, \qquad \log \psi = \log(e^{\mu}) = \mu.$$

Hence $\log T - \log \psi = \bar{X} - \mu$. Because $\bar{X} \sim \text{Normal}(\mu, 1/n)$, we have the *exact* distribution $\sqrt{n}\,(\bar{X} - \mu) \sim \text{Normal}(0, 1)$. Equivalently, define

$$S_{\log}(T, \psi) := \sqrt{n}\,\big(\log T - \log \psi\big) = \sqrt{n}\,(\bar{X} - \mu),$$

then $S_{\log}(T, \psi) \sim \text{Normal}(0, 1)$. This is a pivotal quantity — indeed, an *exact* pivot in this setting. An equivalent form is the ratio-type pivot:

$$\sqrt{n}\,\log(\frac{T}{\psi}) \sim \text{Normal}(0, 1).$$

Since $\xi = T/\psi = e^{\bar{X}-\mu}$ is exactly pivotal (its distribution does not depend on μ), the interval $(T/\xi_{1-\alpha/2}, T/\xi_{\alpha/2})$ — where $\xi_{\alpha/2}$ and $\xi_{1-\alpha/2}$ are estimated from the bootstrap distribution of $T^\star/T$ — can have even better coverage than the interval based on $T - \psi$ (Exercise 4).

Constructing the Pivotal (Empirical Bootstrap) Interval. More generally, consider the pivot $\zeta = T - \psi(F)$, so that $\psi(F) = T - \zeta$. If we knew the $\alpha/2$ and $(1 - \alpha/2)$ quantiles of ζ (say $\zeta_{\alpha/2}$ and $\zeta_{1-\alpha/2}$), then

$$\Pr\big(\zeta_{\alpha/2} \le T - \psi \le \zeta_{1-\alpha/2}\big) = 1 - \alpha \qquad \Longleftrightarrow \qquad \Pr\big(T - \zeta_{1-\alpha/2} \le \psi \le T - \zeta_{\alpha/2}\big) = 1 - \alpha.$$

Of course we do not know these quantiles, since the distribution of ζ depends on the unknown F. The bootstrap approximation replaces F with $\hat{F}$: in the **bootstrap world**, the "true distribution" is $\hat{F}$, so the "true parameter value" is $\psi(\hat{F}) = T$ (e.g., the sample median M from the real data). The bootstrap analog of $\zeta = T - \psi$ is therefore $\zeta^\star = T^\star - T$. We compute $T^\star_b - T$ for each bootstrap replicate, extract the quantiles $\hat{\zeta}_{\alpha/2}$ and $\hat{\zeta}_{1-\alpha/2}$, and plug them in to get

$$\big[\, T - \hat{\zeta}_{1-\alpha/2},\ \ T - \hat{\zeta}_{\alpha/2} \,\big].$$

The same logic applies to any approximately-pivotal quantity. For the ratio pivot $\xi = T/\psi$, we bootstrap $\xi^\star = T^\star/T$, extract its quantiles, and invert to get $(T/\hat{\xi}_{1-\alpha/2}, T/\hat{\xi}_{\alpha/2})$.

Example: the log-ratio pivot for the lognormal. The population pivot is $\xi_{\log} = \sqrt{n}\,\log(T/\psi)$. In the bootstrap world, T plays the role of ψ, so we compute $\xi^\star_{\log,b} = \sqrt{n}\,\log(T^\star_b/T)$ for each replicate and extract the quantiles $q^\star_{\alpha/2}$ and $q^\star_{1-\alpha/2}$. We then invert: starting from

$$\Pr\big(q^\star_{\alpha/2} \ \le\ \sqrt{n}\,\log(\frac{T}{\psi})\ \le\ q^\star_{1-\alpha/2}\big) \approx 1 - \alpha,$$

divide by $\sqrt{n}$, exponentiate, and isolate ψ to obtain the $1 - \alpha$ confidence interval

$$\big[\, T \cdot \exp\big(-\,q^\star_{1-\alpha/2}/\sqrt{n}\big),\ \ T \cdot \exp\big(-\,q^\star_{\alpha/2}/\sqrt{n}\big)\,\big].$$

Note the quantile reversal again: the upper quantile $q^\star_{1-\alpha/2}$ appears in the *lower* bound.

Remark. The progression across Exercises 2–4 illustrates a hierarchy of bootstrap quality: bootstrapping $T^\star$ directly (poor, location-shifted) $\to$ bootstrapping $T^\star - T$ (better, removes location shift but scale still depends on ψ) $\to$ bootstrapping $T^\star/T$ or $\log T^\star - \log T$ (best, fully pivotal). In general, the closer the bootstrapped quantity is to a true pivot, the better the resulting confidence interval.

Remark. Also, we know if we can get a pivotal quantity, which is independent of distribution parameters. The confidence interval may be less sensitive to the data. Because usually, we will use some quantity summarized from data. Non-sensitivity here is referred to as coverage probability, non-sensitive to the true parameter value taken, and no sampling distribution location shifted.

2 GLM Inference

The class of **generalized linear models** assumes that we are working with a dependent variable Y_i that has a distribution in an **exponential dispersion family**.

Exponential Dispersion Family. A family of distributions $\{f(\cdot;\theta,\phi) : \theta \in \Theta, \phi \in \Phi\}$ is an *exponential dispersion family* if we can write

$$f(y;\theta,\phi) = \exp\bigl(\frac{y\theta - b(\theta)}{\phi} + c(y;\phi)\bigr),$$

for some *known* functions $b(\cdot)$ and $c(\cdot,\cdot)$. The parameter θ is the *canonical parameter* and ϕ is the *dispersion parameter*.

Examples of Exponential Dispersion Families Show that the following families are types of exponential dispersion families, and find the corresponding b, c, θ, ϕ

- Y $\sim$ Normal(μ, σ^2)
- $Y = Z/N$ where $Z \sim$ Binomial(N, p)
- $Y \sim$ Poisson(λ)
- $Y \sim$ Gam(α, β)(parameterized so that $\mathbb{E}\bigl[Y\bigr] = \alpha/\beta$)

$$f(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$$

- $Y \sim$ NB(μ, k)

$$f(y \mid \mu, k) = \frac{\Gamma(y+k)}{y!\,\Gamma(k)} \left(\frac{\mu}{\mu+k}\right)^y \left(\frac{k}{\mu+k}\right)^k$$

- $Y \sim$ IG(μ, λ)

$$\begin{aligned} f(y \mid \mu, \lambda) &= \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left[-\frac{\lambda(y-\mu)^2}{2\mu^2 y}\right] \\ &= \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left[-\frac{\lambda y}{2\mu^2} + \frac{\lambda}{\mu} - \frac{\lambda}{2y}\right]. \end{aligned}$$

Generalized Linear Models. Suppose that we have $\mathcal{D} = \{(Y_i, x_i) : i = 1, \dots, N\}$ (with the x_i's regarded as fixed constants). We say that the Y_i's follow a *generalized linear model* if:

1. Y_i has density/mass function

$$f(y_i \mid \theta_i, \phi/\omega_i) = \exp\bigl(\frac{y_i\,\theta_i - b(\theta_i)}{\phi/\omega_i} + c(y_i;\,\phi/\omega_i)\bigr)$$

   where the coefficients $\omega_1, \dots, \omega_N$ are known. This is referred to as the *stochastic component* of the model.

2. For some known (invertible) *link function* $g(\mu)$ we have

$$g(\mu_i) = x_i^\top \beta$$

   where $\mu_i = \mathbb{E}\bigl[Y_i \mid \theta_i, \phi/\omega_i\bigr]$. This is referred to as the *systematic component* of the model. The term $\eta_i = x_i^\top \beta$ is known as the *linear predictor*.

GLM Moments. Suppose that $Y \sim f(y; \theta, \phi/\omega)$ for some exponential dispersion family. The first and second moments are

1. $\mathbb{E}\big[Y \mid \theta, \phi/\omega\big] = b'(\theta)$; and

2. $\mathrm{Var}(Y \mid \theta, \phi/\omega) = \frac{\phi}{\omega} b''(\theta)$.

Variance function

$$\theta_i = (b')^{-1}(\mu_i) = (b')^{-1}\big\{g^{-1}(x_i^\top \beta)\big\}$$

provided that b' and g both have an inverse. Note that GLMs are *heteroskedastic models*, as $\mathrm{Var}(Y \mid \theta, \phi/\omega)$ depends on $\mathbb{E}\big[Y_i \mid \theta, \phi/\omega\big]$. In particular, we have

$$\mathrm{Var}(Y \mid \theta, \phi/\omega) = \frac{\phi}{\omega} b''(\theta) = \frac{\phi}{\omega} b''\big\{(b')^{-1}(\mu)\big\} = \frac{\phi}{\omega} V(\mu).$$

The function $V(\mu) = b''\big\{(b')^{-1}(\mu)\big\}$ is sometimes called the *variance function* of the GLM.
Canonical Link Function. To specify a GLM we must choose the so-called *link function* $g(\mu)$. A convenient choice (for reasons we will discuss later) is $g(\mu) = (b')^{-1}(\mu)$. This is known as the *canonical link*. By definition this gives the model

$$f(y_i \mid x_i, \omega_i, \beta, \phi) = \exp\Big(\frac{y_i x_i^\top \beta - b(x_i^\top \beta)}{\phi/\omega_i} + c(y_i;\, \phi/\omega_i)\Big),$$

i.e., we use the exponential dispersion family with $\theta_i = x_i^\top \beta$. Since $g(\mu) = (b')^{-1}(\mu) = x_i^\top \beta$, and $(b')^{-1}(\mu) = \theta$.

3 Likelihood-Based Inference

GLMs are fit in R using *likelihood based inference*. The likelihood function for a GLM, given data $\mathcal{D} = \{(Y_i, X_i) : i = 1, \dots, N\}$ is given by

$$L(\beta, \phi) = \prod_{i=1}^{N} \exp\Big(\frac{Y_i\theta_i - b(\theta_i)}{\phi/\omega_i} + c(Y_i;\, \phi/\omega_i)\Big),$$

where we define $\theta_i \equiv (b')^{-1}(\mu_i)$ and $\mu_i \equiv g^{-1}(X_i^\top \beta)$. We can then derive the *score function* $s(\beta, \phi) = \frac{\partial}{\partial\beta} \log L(\beta, \phi)$ as

$$s(\beta, \phi) = \sum_{i=1}^{N} \frac{\partial}{\partial\beta} \frac{Y_i\theta_i - b(\theta_i)}{\phi/\omega_i}.$$

Remark. Here, we will write $\frac{\partial}{\partial\beta}F(\beta)$ for the gradient of F and $\frac{\partial^2}{\partial\beta\partial\beta^\top}F(\beta)$ for the Hessian matrix.

Score function. Using the chain rule $\frac{\partial}{\partial\beta} = \frac{\partial}{\partial\theta} \times \frac{\partial\theta}{\partial\mu} \times \frac{\partial\mu}{\partial\beta}$, we have

$$s(\beta, \phi) = \sum_{i=1}^{N} \frac{\omega_i(Y_i - \mu_i)X_i}{\phi V(\mu_i) g'(\mu_i)}.$$

For the canonical link, we have $g'(\mu_i) = V(\mu_i)^{-1}$ so that this reduces to

$$s(\beta, \phi) = \sum_{i=1}^{N} \frac{\omega_i(Y_i - \mu_i)X_i}{\phi}.$$

Remark. Recall that $\frac{d}{dx}g^{-1}(x) = \frac{1}{g'\{g^{-1}(x)\}}$.

Fisher Information. We define the *Fisher Information* to be

$$\mathcal{I}(\beta, \phi) = -\mathbb{E}\big[\frac{\partial^2}{\partial\beta\partial\beta^\top} \log L(\beta, \phi) \mid \beta, \phi\big].$$

The Fisher information plays an important role in inference for GLMs. The "observed" Fisher information is also used,

$$\mathcal{J}(\beta, \phi) = -\frac{\partial^2}{\partial\beta\partial\beta^\top} \log L(\beta, \phi).$$

In addition to being easier to evaluate, using $\mathcal{J}$ has been argued to be the right-thing-to-do™. In any case, show that

$$\langle \mathcal{J}(\beta, \phi)\rangle_{jk} = \frac{1}{\phi}\sum_{i=1}^{N} X_{ij}X_{ik}\left\{\frac{\omega_i}{V(\mu_i)g'(\mu_i)^2} - \frac{\omega_i(Y_i - \mu_i)}{g'(\mu_i)}\left(\frac{\partial}{\partial\mu_i}\frac{1}{V(\mu_i)g'(\mu_i)}\right)\right\}$$

and

$$\langle \mathcal{I}(\beta, \phi)\rangle_{jk} = \frac{1}{\phi}\sum_{i=1}^{N} X_{ij}X_{ik}\frac{\omega_i}{V(\mu_i)g'(\mu_i)^2}$$

where $\langle A\rangle_{ij}$ denotes the $(i,j)^{\text{th}}$ element of the matrix A. Show also that $\mathcal{I}(\beta, \phi) = \mathcal{J}(\beta, \phi)$ when the canonical link is used.

Remark. Notice that the Fisher information has the familiar form

$$\mathcal{I}^{-1} = \phi(\mathbf{X}^\top D \mathbf{X})^{-1}$$

where D is a diagonal matrix with entries $\omega_i / \{V(\mu_i) g'(\mu_i)^2\}$. Similarly, we can write $\mathcal{J}^{-1} = \phi(\mathbf{X}^\top \tilde{D} \mathbf{X})^{-1}$ for some diagonal matrix $\tilde{D}$. Compare this with the linear model, which has inverse Fisher information $\mathcal{I}^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$.

The Basics of Likelihood-Based Inference.

For simplicity, consider data $\mathcal{D} = \{Z_i : i = 1, \ldots, N\}$ with Z_i's iid according to some density $f(z \mid \theta_0)$ where $\{f(\cdot \mid \theta : \theta \in \Theta)\}$ is a family of distributions satisfying some (unstated) regularity conditions. We define the log-likelihood, score, and Fisher information with the equations

$$\ell(\theta) = \sum_{i=1}^N \log f(Z_i \mid \theta), \quad s(\theta) = \frac{\partial}{\partial \theta} \ell(\theta), \quad \text{and} \quad \mathcal{I}(\theta) = -\mathbb{E}\big[\frac{\partial^2}{\partial \theta \partial \theta^\top} \ell(\theta) \mid \theta\big].$$

The following identities are fundamental to likelihood inference:

$$\mathbb{E}\big[s(\theta) \mid \theta\big] = \mathbf{0}, \qquad \text{and} \qquad \operatorname{Var}\{s(\theta) \mid \theta\} = \mathcal{I}(\theta).$$

We will study three types of methods for constructing inference procedures from the likelihood: score methods, Wald methods, and likelihood ratio (LR) methods.

Score Methods. Using the multivariate central limit theorem, show that

$$s(\theta_0) \mathrel{\dot\sim} \text{Normal}\{0, \mathcal{I}(\theta_0)\},$$

but only if we plug in the true value θ_0.

Note: this asymptotic notation means that $X \mathrel{\dot\sim} \text{Normal}(\mu, \Sigma)$ if-and-only-if $\Sigma^{-1/2}(X - \mu) \to \text{Normal}(0, \mathrm{I})$ in distribution.

Wald Methods. Using Taylor's theorem, we have

$$s(\theta_0) = s(\widehat{\theta}) - \mathcal{J}(\theta^\star)(\theta_0 - \widehat{\theta}) = -\mathcal{J}(\theta^\star)(\theta_0 - \widehat{\theta}).$$

where $\theta^\star$ lies on the line segment connecting θ_0 and $\widehat{\theta}$. Now, assume that we know somehow that $\widehat{\theta}$ is a *consistent* estimator of θ_0. Show that

$$\widehat{\theta} \mathrel{\dot\sim} \text{Normal}(\theta_0, \mathcal{I}(\theta_0)^{-1}).$$

Note: you do not need to give a completely rigorous proof. In particular, you can assume that, if $\theta_N \to \theta_0$, then $\frac{1}{N}\mathcal{J}(\theta_N) \to i(\theta_0)$ where

$$i(\theta_0) = -\mathbb{E}\big[\frac{\partial^2}{\partial \theta \partial \theta^\top} \log f(Z_i \mid \theta)\big] = \mathcal{I}(\theta_0)/N.$$

Likelihood Ratio Methods. Consider the Taylor expansion

$$\ell(\theta_0) = \ell(\widehat{\theta}) + s(\widehat{\theta})^\top (\theta_0 - \widehat{\theta}) - \frac{1}{2}(\theta_0 - \widehat{\theta})^\top \mathcal{J}(\theta^\star)(\theta_0 - \widehat{\theta})$$

where $\theta^\star$ lies on the line segment connecting $\widehat{\theta}$ and θ_0. Using Exercise 4, show that

$$-2\{\ell(\theta_0) - \ell(\widehat{\theta})\} \to \chi^2_P.$$

in distribution, where $P = \dim(\theta)$. Recall here that the χ^2_P distribution is the distribution of $\sum_{i=1}^P U_i^2$ where $U_1, \dots, U_P \stackrel{\text{iid}}{\sim} \text{Normal}(0,1)$.

Wilk's Theorem. Suppose that $\{f_{\theta,\eta} : \theta \in \Theta, \eta \in H\}$ is a parametric family satisfying certain regularity conditions. Consider the null hypothesis $H_0 : \eta = \eta_0$, let $\widehat{\theta}_0$ denote the MLE obtained under the null model, and let $(\widehat{\theta}, \widehat{\eta})$ denote the MLE under the unrestricted model. Then, if (θ_0, η_0) denote the values of the parameters that generated the data (so that H_0 is true) then

$$-2\big\{\ell(\widehat{\theta}_0, \eta_0) - \ell(\widehat{\theta}, \widehat{\eta})\big\} \dot\sim \chi^2_D$$

where $D = \dim(\eta)$, as the amount of data tends to ∞.

For Frequentist inference, we can use likelihood-based methods to do things like conduct hypothesis tests. A convenient quantity to use is the *deviance* of a GLM.

Deviance of a GLM. Let $\mathcal{D} = \big\{(Y_i, X_i) : i = 1, \dots, N\big\}$ be modeled with a GLM in the exponential dispersion family with canonical parameters $\theta_i = (b')^{-1}(\mu_i) = (b')^{-1}(g^{-1}(X_i^\top \beta))$. Let $\theta = (\theta_1, \dots, \theta_N)$ and let $\widehat{\theta} = (\widehat{\theta}_1, \dots, \widehat{\theta}_N)$ be the MLE of the θ's under our model. We define the *saturated model* as the model which has a separate parameter for all unique values of x in $\mathcal{D}$:

$$f(y \mid x, \phi/\omega) = \exp\Big(\frac{y\theta_x - b(\theta_x)}{\phi/\omega} + c(y;\, \phi/\omega)\Big).$$

The **residual deviance** of a model is defined by

$$D = -2\phi\big\{\ell(\widehat{\theta}) - \ell(\widehat{\theta}_x)\big\}$$

where $\ell(\theta) = \sum_{i=1}^N \frac{\omega_i(Y_i\theta_i - b(\theta_i))}{\phi}$ is the log-likelihood of θ and $\widehat{\theta}_{xi} = (b')^{-1}(Y_i)$. The **scaled deviance** is $D^\star = D/\phi$.

Note: if ϕ is unknown, we generally replace ϕ with an estimate $\widehat{\phi}$ given by

$$\widehat{\phi} = \frac{1}{N-P}\sum_{i=1}^N \frac{\omega_i(Y_i - \widehat{\mu}_i)^2}{V(\widehat{\mu}_i)}.$$

Estimating the Dispersion. Show that the quantity

$$\widetilde{\phi} = \frac{1}{N}\sum_i \frac{\omega_i(Y_i - \mu_i)^2}{V(\mu_i)}$$

is unbiased for ϕ. We don't use $\widetilde{\phi}$ because we don't know the μ_i's, so the modified denominator in $\widehat{\phi}$ compensates for the "degrees of freedom" used to estimate β.

In words: the scaled deviance is the LRT statistic for comparing the model with the saturated model which has the maximal number of model parameters in the GLM.

4 GLMs

Poisson Regression (Poisson Log-Linear Model).

Poisson Regression $Y_i \sim \text{Poisson}(\mu_i)$, where $\log(\mu_i) = x_i^\top \beta$. Equivalently, we have $\mu_i = \exp\bigl(x_i^\top \beta\bigr)$.
Interpretation. Suppose we fit a Poisson log-linear model $\log(\mu_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2}$. A change in X_{i2} by δ units, holding X_{i1} fixed, results in a multiplicative effect on the mean:

$$\mu_{\text{new}} = e^{\beta_2 \delta}\, \mu_{\text{old}}$$

Issue. A problem with Poisson log-linear models is that they impose the restriction $\mathbb{E}\bigl[Y_i\bigr] = \text{Var}(Y_i)$ so that the variance is completely constrained by the mean. Count data is referred to as *overdispersed* if $\text{Var}(Y_i) > \mathbb{E}\bigl[Y_i\bigr]$.

Logistic Regression

Logistic Regression. A particular case of a GLM takes $Y_i = Z_i/n_i$ where $Z_i \sim \text{Binomial}(n_i, p_i)$. When used with the canonical link $g(p) = \text{logit}(p) = \log \frac{p}{1-p}$, we arrive at the *logistic regression model*:

$$p_i = \frac{\exp\bigl(x_i^\top \beta\bigr)}{1 + \exp\bigl(x_i^\top \beta\bigr)} \quad \text{equivalently} \quad \text{logit}(p_i) = x_i^\top \beta.$$

This model is used to model outcomes that correspond to counts from a binomial experiment (i.e., repeatedly flipping a coin n_i times with probability p_i of heads, and counting the number of heads). The special case $n_i = 1$ is very common, in which case the outcomes Y_i are binary.
Likelihood. Since $Z_i \sim \text{Binomial}(n_i, p_i)$ is the random variable with a known distribution (Y_i's distribution unknown), we write the likelihood directly from its PMF:

$$L(\beta) = \prod_{i=1}^{N} \binom{n_i}{Z_i} p_i^{Z_i} (1 - p_i)^{n_i - Z_i}.$$

Taking logarithms and dropping the combinatorial constant (which does not depend on β) gives the log-likelihood

$$\ell(\beta) = \sum_{i=1}^{N} \bigl[Z_i \log p_i + (n_i - Z_i) \log(1 - p_i)\bigr] = \sum_{i=1}^{N} \bigl[Z_i\, x_i^\top \beta - n_i \log\bigl(1 + \exp\bigl(x_i^\top \beta\bigr)\bigr)\bigr],$$

where the second equality follows from substituting $\log p_i = x_i^\top \beta - \log\bigl(1 + \exp\bigl(x_i^\top \beta\bigr)\bigr)$ and $\log(1 - p_i) = -\log\bigl(1 + \exp\bigl(x_i^\top \beta\bigr)\bigr)$. For the common case $n_i = 1$ (binary outcomes $Y_i \in \{0, 1\}$), we have $Z_i = Y_i$ and this simplifies to

$$\ell(\beta) = \sum_{i=1}^{N} \bigl[Y_i\, x_i^\top \beta - \log\bigl(1 + \exp\bigl(x_i^\top \beta\bigr)\bigr)\bigr].$$

Why define Y_i? The variable $Y_i = Z_i/n_i$ is introduced for convenience, not for the likelihood: $Y_i \in [0, 1]$ is always a proportion (comparable across observations with different n_i), and $\mathbb{E}\bigl[Y_i\bigr] = p_i$ lets us write the link cleanly as $\text{logit}\bigl(\mathbb{E}\bigl[Y_i\bigr]\bigr) = x_i^\top \beta$.

Odds. The *odds* of success is defined as $\text{Odds}(p) = \frac{p}{1-p}$, so that $\text{logit}(p) = \log \text{Odds}(p)$.

good luck :)

Interpretation. Suppose we fit a logistic regression model $\text{logit}(p_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2}$. Logistic regression models are often interpreted in terms of *odds ratios*; the odds of success of observational unit i relative to i' is given by

$$\frac{\text{Odds}(Y_i = 1 \mid X_i)}{\text{Odds}(Y_{i'} = 1 \mid X_{i'})} = \frac{\Pr(Y_i = 1 \mid X_i)\,\Pr(Y_{i'} = 0 \mid X_{i'})}{\Pr(Y_i = 0 \mid X_i)\,\Pr(Y_{i'} = 1 \mid X_{i'})}.$$

If X_i and $X_{i'}$ are identical except that $X_{i2} = X_{i'2} + \delta$, then the odds ratio is $e^{\beta_2\delta}$. That is, *shifting a covariate by* δ has a multiplicative effect on the odds of success, inflating the odds by a factor of $e^{\beta_2\delta}$.

Alternative Link Functions. While the logit link is the most popular link used in practice, it is not the only commonly used link function. Different link functions can all be understood in terms of the following latent variable formulation of binomial regression. We consider a continuous latent (unobserved) variable Z_i and a binary outcome of interest Y_i such that

$$Y_i = \mathbf{1}(Z_i > 0).$$

We then assume that $Z_i = X_i^\top\beta + \epsilon_i$ where ϵ_i follows some distribution with cdf F to derive an implicit model for the Y_i's.
Outcome Model. The model for Y_i this leads to is

$$\Pr(Y_i = 1 \mid X_i = x) = 1 - F(-x^\top\beta).$$

Further, if $F(\cdot)$ is the cdf of a symmetric distribution, then $\Pr(Y_i = 1 \mid X_i = x) = F(x^\top\beta)$.
Different Links from Different ϵ_i. Different choices of ϵ_i lead to different link functions. When F is symmetric, $g = F^{-1}$ directly; when F is not symmetric, we must work from the general form $p = 1 - F(-x^\top\beta)$.

- $\epsilon_i \sim \text{Logistic}(0,1)$ (symmetric): F is the logistic cdf, $g(p) = F^{-1}(p) = \log\frac{p}{1-p} \Rightarrow$ **logit link**.
- $\epsilon_i \sim \text{Normal}(0,1)$ (symmetric): $F = \Phi$, $g(p) = \Phi^{-1}(p) \Rightarrow$ **probit link**.
- $\epsilon_i \sim t_\nu$ (symmetric): $g(p) = F_{t_\nu}^{-1}(p) \Rightarrow$ **robit link** (a "robust" probit/logit). The robit link with $\nu = 7$ is a good approximation to the logit link.
- $\epsilon_i \sim$ Gumbel (not symmetric): leads to the **log-log link**, $g(p) = -\log(-\log p)$.
- $-\epsilon_i \sim$ Gumbel (not symmetric): leads to the **complementary log-log link**, $g(p) = \log(-\log(1-p))$.

Derivation: Gumbel $\Rightarrow$ log-log link. Let $\epsilon_i \sim$ Gumbel with cdf $F(\epsilon) = 1 - e^{-e^{\epsilon}}$. Since Gumbel is not symmetric, we use the general form. Let $\eta = x^\top\beta$:

$$p = 1 - F(-\eta) = 1 - \left(1 - e^{-e^{-\eta}}\right) = e^{-e^{-\eta}}.$$

Inverting: $\log p = -e^{-\eta}$, then $-\log p = e^{-\eta}$, then $\log(-\log p) = -\eta$, so

$$g(p) = -\log(-\log p) = \eta = x^\top\beta.$$

Derivation: $-\epsilon_i \sim$ Gumbel $\Rightarrow$ cloglog link. If $-\epsilon_i \sim$ Gumbel, then $F_\epsilon(t) = 1 - F_{\text{Gumbel}}(-t) = e^{-e^{-t}}$. Again let $\eta = x^\top\beta$:

$$p = 1 - F_\epsilon(-\eta) = 1 - e^{-e^{\eta}}.$$

Inverting: $1 - p = e^{-e^{\eta}}$, then $-\log(1-p) = e^{\eta}$, then $\log(-\log(1-p)) = \eta$, so

$$g(p) = \log(-\log(1-p)) = \eta = x^\top\beta.$$

Logistic Regression in Case-Control Studies. In addition to being associated to the canonical link (HW3 Q8 (c)), another reason that a lot of people (especially in medicine) like to use logistic regression is due to its special relationship with *case-control studies*.

Case-control study is a particular sampling design that over-represents individuals experiencing an outcome of interest (cases) relative to those who do not (controls). For example, to study the impact of smoking on healthcare, the natural thing to do is to randomly sample from the population of smokers and non-smokers and see if the rates of lung cancer differ; case-control studies **do the opposite** and instead sample a group of people with lung cancer and a group of people without lung cancer and see if the rates of smoking differ between the two groups. This can be useful for a lot of reasons, but one is that lung cancer is rare in general, and a random sample of smokers/non-smokers might have very few individuals with lung cancer.

Specifically, in the population, let $Y_i = 1$ denote a *case*, $Y_i = 0$ denote a *control*, and X_i denote some covariates of interest. Scientifically, we might want to estimate a model:

$$\Pr(Y_i = 1 \mid X_i = x) = \frac{e^{\alpha + x^\top \beta}}{1 + e^{\alpha + x^\top \beta}},$$

where α is an intercept term and β is a vector of regression coefficients. Now, suppose that *by design* we sample from the population so that individual i is selected with probability π_1 if they are a case and π_0 if they are a control. Let $S_i = 1$ if the individual is sampled and $S_i = 0$ otherwise. Then, assuming $\Pr(Y_i = 1 \mid X_i = x)$ is the logistic form above,

$$\Pr(Y_i = 1 \mid X_i = x, S_i = 1) = \frac{e^{\alpha^\star + x^\top \beta}}{1 + e^{\alpha^\star + x^\top \beta}}$$

for some $\alpha^\star \neq \alpha$, where $\alpha^\star = \alpha + \log(\pi_1/\pi_0)$. That is, case-control sampling only changes the intercept; the coefficients β remain the same. **What is the implication for how we can use case-control studies to learn β?**

<u>**Remark.**</u> The difference between a randomized clinical trial and a case-control study is the sampling criteria. The RCT is sampling data based on a binary covariate X_i, while the case-control study is sampling data based on the response variable Y_i.

5 Overdispersion

Overdispersion overview

Overdispersion is a common phenomenon for count/binomial type data. By "binomial type" data, we mean the usual setting of repeated success/failure experiments with each experiment having success probability p, but possibly without the assumption of independence between trials and with only the number of successes reported.
Overdispersion in GLM Modelling. For GLMs we know that $\mathrm{Var}(Y_i \mid X_i) = \frac{\phi}{\omega_i} V(\mu_i)$, so that ϕ is linked directly to the variability in Y_i. For the Poisson and Binomial GLMs, we know exactly that $\phi \equiv 1$; overdispersion relative to these models occurs when $\mathrm{Var}(Y_i \mid X_i) > \omega_i^{-1} V(\mu_i)$.
Overdispersion in GLM Inference. The problem introduced by overdispersion is primarily that it affects our uncertainty quantification for the regression coefficients. The asymptotic variance of the regression coefficients in a GLM is given by

$$\mathrm{Var}(\widehat{\beta}) \approx \phi (X^\top W X)^{-1},$$

where W depends on the means μ_i and the weights ω_i, but notably not on ϕ. If the self-imposed value $\phi = 1$ is too small, we will end up underestimating $\mathrm{Var}(\widehat{\beta})$. This implies that failure to account for overdispersion has a material consequence: we will end up *underestimating our uncertainty in* β! This will cause (for example) our confidence intervals to not have nominal coverage levels and our hypothesis tests to have higher than nominal error rates.

Dealing with Overdispersion. The most obvious strategy for dealing with overdispersion is to use a larger (but still parametric) probabilistic model that can accommodate it. The simplest models to use are the **negative binomial** model for count data and the **beta-binomial** model for binomial-type data.

Negative Binomial Regression

Negative Binomial Regression. The negative binomial model sets $[Y_i \mid \mu_i, k] \sim f(y \mid \mu_i, k)$ where $f(y \mid \mu, k)$ is a negative binomial mass function

$$f(y \mid \mu, k) = \frac{\Gamma(y+k)}{y!\,\Gamma(k)} \left(\frac{\mu}{\mu+k}\right)^y \left(\frac{k}{\mu+k}\right)^k.$$

The link function is typically taken to be the log link: $\log(\mu_i) = x_i^\top \beta$, consistent with Poisson regression.
NB as Exponential Dispersion Family. Define $\theta = \log \frac{\mu}{\mu+k}$, so that $e^\theta = \frac{\mu}{\mu+k}$ and $\frac{k}{\mu+k} = 1 - e^\theta$. Then

$$\log f(y \mid \mu, k) = y\theta - \left[-k \log(1 - e^\theta)\right] + \log \frac{\Gamma(y+k)}{y!\,\Gamma(k)} = \frac{y\theta - b(\theta)}{1} + c(y, k),$$

with $b(\theta) = -k \log(1 - e^\theta)$, $\phi = 1$, and $c(y,k) = \log \frac{\Gamma(y+k)}{y!\,\Gamma(k)}$. This is exactly the exponential dispersion family form with $\phi = 1$.

k vs ϕ. The variance of the negative binomial is $\mathrm{Var}(Y) = \mu + \mu^2/k$, which consists of the Poisson variance μ plus an overdispersion term μ^2/k. As $k \to \infty$, $\mathrm{Var}(Y) \to \mu$ and we recover the Poisson model. Hence k is better understood as a **shape parameter** that governs how quickly the variance grows with the mean: it alters the functional form of the variance rather than acting as a multiplicative scale factor like ϕ.

Beta-Binomial Regression. Binomial-type data Z with mean np is overdispersed if $\text{Var}(Z) > np(1-p)$. Binomial regression assumes $Z_i \sim \text{Binomial}(n_i, p_i)$, which requires *independence* across trials. In many cases we won't have strong evidence for this assumption; the beta-binomial model provides an analogue of the negative binomial for handling overdispersion in binomial-type data.

Beta-Binomial Distribution

Beta-Binomial Distribution. Suppose $Z_i \sim \text{Binomial}(n_i, p_i)$ with $p_i \sim \text{Beta}\big\{\rho\mu_i,\, \rho(1-\mu_i)\big\}$. Marginally, Z_i has mass function

$$f(z;\, \mu_i, \rho) = \binom{n_i}{z} \cdot \frac{\Gamma(\rho)}{\Gamma(\rho\mu_i)\,\Gamma(\rho[1-\mu_i])} \cdot \frac{\Gamma(\rho\mu_i + z)\,\Gamma(\rho[1-\mu_i] + n_i - z)}{\Gamma(\rho + n_i)}.$$

This is known as the *beta-binomial distribution*, with $\mathbb{E}\big[Z_i\big] = n_i\mu_i$.
Overdispersion. By the law of total variance:

$$\text{Var}(Z_i) = \mathbb{E}\big[\text{Var}(Z_i \mid p_i)\big] + \text{Var}(\mathbb{E}\big[Z_i \mid p_i\big]).$$

Since $Z_i \mid p_i \sim \text{Binomial}(n_i, p_i)$, the within-group term is $\mathbb{E}\big[n_i p_i(1-p_i)\big] = n_i(\mu_i(1-\mu_i) - \text{Var}(p_i))$, and the between-group term is $\text{Var}(n_i p_i) = n_i^2\,\text{Var}(p_i)$. Summing:

$$\text{Var}(Z_i) = n_i\mu_i(1-\mu_i) + n_i(n_i - 1)\,\text{Var}(p_i).$$

Since $n_i > 1$ and $\text{Var}(p_i) > 0$, we have $\text{Var}(Z_i) > n_i\mu_i(1-\mu_i)$, so Z_i is **overdispersed** relative to the $\text{Binomial}(n_i, \mu_i)$ distribution.
Remark. The proof above never requires the explicit form of the Beta distribution — only that $\text{Var}(p_i) > 0$. This shows that overdispersion is a universal consequence of **any** random-effects model in which the success probability varies across observations, regardless of the distributional assumption on p_i.

Zero-Inflation

Another common issue with count and binomial data is that it is often the case that there are more zero counts than would be expected if the data were actually binomial/Poisson.
Zero-inflation can occur in many practical settings where two distinct processes are generating the data: one process determining whether an event can occur at all, and another process determining the count when events are possible.

Zero-Inflated Poisson (ZIP)

To deal with zero-inflation, a common strategy is to model the probability of a zero as being the probability of a *structural zero* plus the probability of a zero from the Poisson distribution. The ZIP model sets

$$\Pr(Y_i = y \mid X_i) = \begin{cases} \pi_i + (1-\pi_i)e^{-\mu_i} & \text{if } y_i = 0 \\ (1-\pi_i)\dfrac{e^{-\mu_i}\mu_i^{y_i}}{y_i!} & \text{if } y_i > 0 \end{cases}$$

This models Y_i as a mixture of two distributions: (1) a point-mass at 0 with probability π_i, and (2) a Poisson distribution with mean μ_i with probability $1 - \pi_i$. Both parameters can be modeled using covariates, e.g.,

$$\text{logit}(\pi_i) = X_i^\top \beta_\pi, \qquad \text{and} \qquad \log(\mu_i) = X_i^\top \beta_\mu.$$

Remark. The ZIP model also induces overdispersion. By the law of total variance, writing Y_i as a mixture with probability π_i on a point mass at 0 and probability $1-\pi_i$ on $\text{Pois}(\mu_i)$:

$$\text{Var}(Y_i) = \mathbb{E}\big[\text{Var}(Y_i \mid \lambda_i)\big] + \text{Var}(\mathbb{E}\big[Y_i \mid \lambda_i\big]) = \mathbb{E}\big[\lambda_i\big] + \text{Var}(\lambda_i) > \mathbb{E}\big[\lambda_i\big] = \mathbb{E}\big[Y_i\big] = (1-\pi_i)\mu_i.$$

Hence $\text{Var}(Y_i) > \mathbb{E}\big[Y_i\big]$, so the ZIP model is automatically overdispersed relative to the Poisson. This is consistent with the general principle: any mixture or random-effects model that introduces variability in the rate parameter will produce overdispersion.

Hurdle Model

A seemingly-equivalent model to the ZIP model is the *hurdle model*, which instead takes

$$Y_i \sim \pi_i\,\delta_0 + (1-\pi_i)\,\text{TruncPois}(\mu_i)$$

where $\text{logit}(\pi_i) = X_i^\top \beta_\pi$ and TruncPois denotes the Poisson distributed truncated to $\{1, 2, \ldots\}$ (i.e., excluding zero) with (non-truncated) mean $\mu_i = e^{X_i^\top \beta_\mu}$. Advantages of this approach include: (1) the likelihood factors as $L(\beta_\pi, \beta_\mu) = L(\beta_\pi)\,L(\beta_\mu)$, so parameters can be estimated by fitting a logistic regression to $Z_i = \mathbf{1}(Y_i = 0)$ and a $\text{TruncPois}(e^{X_i^\top \beta_\mu})$ to all observations with $Y_i > 0$; (2) it allows for *zero deflation* (i.e., fewer zeros than implied by the Poisson), which the ZIP model does not. A downside is that the model no longer distinguishes between structural and non-structural zeros.

Remark. The likelihood of the hurdle model for observation i can be written as

$$L_i(\beta_\pi, \beta_\mu) = \begin{cases} \pi_i & \text{if } Y_i = 0 \\ (1-\pi_i)\cdot f_{\text{TP}}(Y_i;\, \mu_i) & \text{if } Y_i > 0 \end{cases}$$

where f_{TP} is the truncated Poisson mass function. The full likelihood is then

$$L(\beta_\pi, \beta_\mu) = \prod_{i:\,Y_i=0} \pi_i \prod_{i:\,Y_i>0} (1-\pi_i)\cdot f_{\text{TP}}(Y_i;\, \mu_i) = \underbrace{\prod_{i=1}^{N} \pi_i^{Z_i}(1-\pi_i)^{1-Z_i}}_{L(\beta_\pi)} \cdot \underbrace{\prod_{i:\,Y_i>0} f_{\text{TP}}(Y_i;\, \mu_i)}_{L(\beta_\mu)}$$

where $Z_i = \mathbf{1}(Y_i = 0)$. Since $L(\beta_\pi)$ depends only on β_π and $L(\beta_\mu)$ depends only on β_μ, maximizing the joint likelihood is equivalent to maximizing each part separately: a logistic regression on Z_i for β_π, and a truncated Poisson regression on the observations with $Y_i > 0$ for β_μ. In contrast, the ZIP likelihood is

$$L(\beta_\pi, \beta_\mu) = \prod_{i:\,Y_i=0} \big[\pi_i + (1-\pi_i)e^{-\mu_i}\big] \prod_{i:\,Y_i>0} (1-\pi_i)\frac{e^{-\mu_i}\mu_i^{Y_i}}{Y_i!}$$

which *cannot* be factored this way: when $Y_i = 0$, the contribution $\pi_i + (1-\pi_i)e^{-\mu_i}$ couples β_π and β_μ together additively, so they must be estimated jointly.

Zero-Inflated Negative Binomial (ZINB)

The ZINB model is analogous to the ZIP model, except that instead of mixing a point mass at zero with a Poisson, we mix a point mass at zero with a Negative Binomial distribution. Specifically,

$$\Pr(Y_i = y \mid X_i) = \begin{cases} \pi_i + (1-\pi_i)\,f_{\text{NB}}(y;\, \mu_i, k)\big|_{y=0} & \text{if } y = 0, \\ (1-\pi_i)\,f_{\text{NB}}(y;\, \mu_i, k) & \text{if } y > 0, \end{cases}$$

where $\pi_i = \Pr(\text{structural zero})$ is modeled via logistic regression (e.g. $\operatorname{logit}(\pi_i) = X_i^\top \beta_\pi$), f_{NB} is the negative binomial mass function with mean μ_i and dispersion parameter k, and μ_i is modeled with the log link (e.g. $\log \mu_i = X_i^\top \beta_\mu$). This model is useful when (i) there are many structural zeros and (ii) even ignoring these structural zeros, the count data is overdispersed.

Key insight. (cf. Exercise 6) When comparing ZIP, ZINB, NB, and Poisson models, the ZIP model may produce misleadingly narrow confidence intervals. This is because ZIP handles zero-inflation but its count component is still Poisson, which imposes $\operatorname{Var}(Y) = \mathbb{E}[Y]$. If the non-zero counts are overdispersed, the Poisson component will underestimate the variance, leading to understated standard errors and overconfident inference. The ZINB model addresses both issues simultaneously: π_i captures structural zeros while k accommodates overdispersion in the count component, yielding wider but more honest confidence intervals.

6 Vector GLMs

Vector Exponential Dispersion Family

A family of distributions is a *vector exponential dispersion family* if we can write

$$f(y;\,\theta,\phi) = \exp\bigl(\frac{y^\top\theta - b(\theta)}{\phi} + c(y,\phi)\bigr)$$

where $y = (y_1, \ldots, y_K)^\top$ and $\theta = (\theta_1, \ldots, \theta_K)^\top$. In component form:

$$f(y;\,\theta,\phi) = \exp\bigl(\frac{\sum_{i=1}^K y_i\theta_i - b(\theta)}{\phi} + c(y,\phi)\bigr).$$

The moments generalize the scalar case:

$$\mathbb{E}\bigl[Y\bigr] = \nabla b(\theta), \qquad \mathrm{Var}(Y) = \phi\,\nabla^2 b(\theta),$$

i.e., $\mathbb{E}\bigl[Y_i\bigr] = \frac{\partial b(\theta)}{\partial\theta_i}$ and $\mathrm{Cov}(Y_i, Y_j) = \phi\,\frac{\partial^2 b(\theta)}{\partial\theta_i\,\partial\theta_j}$.

Multinomial Logit Model

For nominal outcomes with K categories, we model the outcome Y_i using a multinomial distribution:

$$[Y_i \mid X_i = x] \sim \mathrm{Multinomial}\bigl\{n;\,\pi_1(x), \ldots, \pi_K(x)\bigr\}$$

where n is the number of samples taken on Y_i and $\pi_k(x)$ is the probability of observing the k^{th} category; that is, Y_i has mass function

$$\Pr(Y_i = y \mid X_i = x) = \binom{n}{y_1, y_2, \ldots, y_K} \prod_{k=1}^K \pi_k(x)^{y_k}$$

for all vectors of non-negative integers $(y_1, \ldots, y_K)$ with $\sum_k y_k = n$. Our goal is to model these probabilities as functions of X_i while ensuring $0 \le \pi_k(x) \le 1$ for all k, x and $\sum_k \pi_k(x) = 1$ for all x. The *multinomial logit model* achieves this by setting:

$$\pi_k(x) = \frac{\exp\bigl(x^\top\beta_k\bigr)}{\sum_j \exp\bigl(x^\top\beta_j\bigr)}.$$

The parameter of interest is the $K \times P$ matrix β with rows $\beta_1, \ldots, \beta_K$, so that $\beta x = \eta = (\eta_1, \ldots, \eta_K)^\top$ returns a $K \times 1$ vector of linear predictors.

Likelihood

Given data $\mathcal{D} = \bigl\{(Y_i, X_i) : i = 1, \ldots, N\bigr\}$ where $Y_i = (Y_{i1}, \ldots, Y_{iK})^\top$ with $\sum_k Y_{ik} = n_i$, the likelihood for the multinomial logit model is

$$L(\beta) = \prod_{i=1}^N \binom{n_i}{Y_{i1}, \ldots, Y_{iK}} \prod_{k=1}^K \pi_k(x_i)^{Y_{ik}} = \prod_{i=1}^N \binom{n_i}{Y_{i1}, \ldots, Y_{iK}} \prod_{k=1}^K \Bigl(\frac{\exp\bigl(x_i^\top\beta_k\bigr)}{\sum_j \exp\bigl(x_i^\top\beta_j\bigr)}\Bigr)^{Y_{ik}}.$$

For the common case $n_i = 1$ (each observation falls into exactly one category), Y_i is a one-hot vector with $Y_{ic_i} = 1$ for the observed category c_i and $Y_{ik} = 0$ for $k \neq c_i$. The multinomial coefficient equals 1 and the likelihood simplifies to

$$L(\beta) = \prod_{i=1}^{N} \pi_{c_i}(x_i) = \prod_{i=1}^{N} \frac{\exp\bigl(x_i^\top \beta_{c_i}\bigr)}{\sum_j \exp\bigl(x_i^\top \beta_j\bigr)}.$$

Log-Likelihood. Taking logs, the general log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{N} \Bigl[\sum_{k=1}^{K} Y_{ik}\, x_i^\top \beta_k - n_i \log \sum_{j=1}^{K} \exp\bigl(x_i^\top \beta_j\bigr)\Bigr] + \text{const}$$

where the constant is $\sum_i \log \binom{n_i}{Y_{i1},\ldots,Y_{iK}}$, which does not depend on β. For the case $n_i = 1$, this simplifies to

$$\ell(\beta) = \sum_{i=1}^{N} \bigl[x_i^\top \beta_{c_i} - \log \sum_{j=1}^{K} \exp\bigl(x_i^\top \beta_j\bigr)\bigr].$$

This is recognizable as the *cross-entropy loss* (up to a sign) commonly used in the machine learning literature.

Remark. Under the baseline category identification $\beta_K = 0$, the log-sum-exp term becomes $\log\bigl(1 + \sum_{j=1}^{K-1} \exp\bigl(x_i^\top \beta_j\bigr)\bigr)$. When $K = 2$, this reduces to

$$\ell(\beta_1) = \sum_{i=1}^{N} \bigl[Y_i\, x_i^\top \beta_1 - \log\bigl(1 + \exp\bigl(x_i^\top \beta_1\bigr)\bigr)\bigr]$$

which is exactly the binary logistic regression log-likelihood derived in Section 4.

Remark. In multinomial GLMs, the link function is applied to the category probabilities π_k, not directly to the counts Y_{ik}. Although $\mathbb{E}\bigl[Y_{ik}\bigr] = n_i \pi_k$, the ratio structure of the logit link eliminates the dependence on n_i (since $\frac{\mathbb{E}[Y_{ik}]}{\mathbb{E}[Y_{iK}]} = \frac{n_i \pi_k}{n_i \pi_K} = \frac{\pi_k}{\pi_K}$), so no explicit scaling is required.

Identifiability

Identifiability Issue The multinomial logit model is not identified: for any β, replacing β_k with $\beta_k' = \beta_k + c$ for all k yields $L(\beta) = L(\beta')$ with $\beta \neq \beta'$, since $e^{x^\top c}$ cancels in the ratio $\pi_k(x) = e^{x^\top \beta_k} / \sum_j e^{x^\top \beta_j}$. Hence there exist infinitely many β' with the same likelihood, and further constraints are needed.

Identification Strategies. Two common approaches:

- **Baseline category logits:** Set $\beta_K = 0$ for some reference category K. This gives $\log \frac{\pi_k(x)}{\pi_K(x)} = x^\top \beta_k$ for $k = 1, \ldots, K-1$, allowing clear interpretation of coefficients as log-odds ratios relative to the baseline.
- **Symmetric constraints:** Set $\sum_k \beta_k = 0$. This treats all categories symmetrically, which is preferable when using regularization or informative priors, as no single category is exempt from penalization.

Interpretation

Under the baseline category logit with $\beta_K = 0$, conditioning on $Y_i \in \{k, K\}$, define

$$p = \Pr(Y_i = k \mid X_i = x,\, Y_i \in \{k, K\}) = \frac{\pi_k(x)}{\pi_k(x) + \pi_K(x)}, \qquad 1 - p = \frac{\pi_K(x)}{\pi_k(x) + \pi_K(x)}.$$

These sum to 1, so the classical $p/(1-p)$ definition applies. The scaling factor $\pi_k(x) + \pi_K(x)$ cancels, giving

$$\mathrm{Odds}(Y_i = k \mid X_i = x,\, Y_i \in \{k, K\}) = \frac{p}{1-p} = \frac{\pi_k(x)}{\pi_K(x)} = \exp\!\big(x^\top \beta_k\big).$$

If x and x' are identical except that $x_j = x'_j + 1$, then $e^{\beta_{kj}}$ corresponds to a multiplicative effect on the odds ratio:

$$\frac{\mathrm{Odds}(Y_i = k \mid X_i = x,\, Y_i \in \{k, K\})}{\mathrm{Odds}(Y_i = k \mid X_i = x',\, Y_i \in \{k, K\})} = \frac{\exp\!\big(x^\top \beta_k\big)}{\exp\!\big(x'^\top \beta_k\big)} = e^{\beta_{kj}}.$$

Furthermore, since $\beta_K = 0$ we have $\pi_K(x) \propto 1$, so $p = \frac{\exp\left(x^\top \beta_k\right)}{\exp\left(x^\top \beta_k\right) + 1} = \sigma(x^\top \beta_k)$ where $\sigma(\cdot)$ is the logistic (sigmoid) function. This reveals that the multinomial logit, restricted to $\{k, K\}$, reduces to a standard binary logistic regression. Since $\sigma(\cdot)$ is strictly monotone increasing:

- $\beta_{kj} > 0 \Rightarrow e^{\beta_{kj}} > 1$: increasing x_j by 1 **increases** the odds and probability of category k relative to K; $\mathrm{odds}_{\mathrm{new}} > \mathrm{odds}_{\mathrm{old}}$, $p_{\mathrm{new}} > p_{\mathrm{old}}$.
- $\beta_{kj} < 0 \Rightarrow e^{\beta_{kj}} < 1$: increasing x_j by 1 **decreases** the odds and probability of category k relative to K; $\mathrm{odds}_{\mathrm{new}} < \mathrm{odds}_{\mathrm{old}}$, $p_{\mathrm{new}} < p_{\mathrm{old}}$.
- $\beta_{kj} = 0 \Rightarrow e^{\beta_{kj}} = 1$: predictor j has **no effect** on the odds or probability of category k versus K; $\mathrm{odds}_{\mathrm{new}} = \mathrm{odds}_{\mathrm{old}}$, $p_{\mathrm{new}} = p_{\mathrm{old}}$.

Cumulative Link Ordinal Models

Motivation: Latent Variable Formulation. For ordinal outcomes, we think of the observed categories as a discretization of an underlying continuous latent variable. We suppose that there exists a latent variable Z_i and unknown threshold parameters $\alpha_1 < \alpha_2 < \cdots < \alpha_{K-1}$ such that:

- If $Z_i \le \alpha_1$, we observe $Y_i = 1$.
- If $\alpha_{k-1} < Z_i \le \alpha_k$, we observe $Y_i = k$, for $k = 2, \ldots, K-1$.
- If $Z_i > \alpha_{K-1}$, we observe $Y_i = K$.

If we assume $Z_i = x^\top \beta + \epsilon$ with $\epsilon \sim \mathrm{Normal}(0, 1)$, then we can derive the cumulative probabilities:

$$\begin{aligned}
\Pr(Y_i \le k \mid X_i = x) &= \Pr(Z_i \le \alpha_k \mid X_i = x) = \Pr(x^\top \beta + \epsilon \le \alpha_k) \\
&= \Pr(\epsilon \le \alpha_k - x^\top \beta) = \Phi(\alpha_k - x^\top \beta)
\end{aligned}$$

where $\Phi(\cdot)$ is the standard normal CDF.

Link Functions

The cumulative link model can be written in a unified form as:

$$g(\Pr(Y_i \le k \mid X_i = x)) = \alpha_k - x^\top \beta$$

where g is a link function applied to the cumulative probabilities. Different choices of g correspond to different assumptions on the distribution of the latent error ϵ in $Z_i = x^\top \beta + \epsilon$:

- **Probit link:** $\epsilon \sim \text{Normal}(0,1)$, so $g(p) = \Phi^{-1}(p)$ and $\Pr(Y_i \le k \mid x) = \Phi(\alpha_k - x^\top \beta)$.
- **Logit link:** $\epsilon \sim \text{Logistic}(0,1)$, so $g(p) = \log \frac{p}{1-p}$ and $\Pr(Y_i \le k \mid x) = \frac{e^{\alpha_k - x^\top \beta}}{1+e^{\alpha_k - x^\top \beta}}$.
- **Complementary log-log link:** $-\epsilon \sim \text{Gumbel}$, so $g(p) = \log(-\log(1-p))$ and $\Pr(Y_i \le k \mid x) = 1 - \exp(-\exp(\alpha_k - x^\top \beta))$.

The logit link gives the *cumulative logit model* (sometimes called the *proportional odds logistic regression* model). Note that the odds here is defined on *cumulative* probabilities — the odds of being in category k or lower versus being above k:

$$\log\left(\frac{\Pr(Y_i \le k \mid X_i = x)}{\Pr(Y_i > k \mid X_i = x)}\right) = \log\left\{\text{Odds}(Y_i \le k \mid X_i = x)\right\} = \alpha_k - x^\top \beta.$$

Remark. With $\epsilon \sim \text{Normal}(0,1)$, the model above is referred to as the *cumulative probit model.* If we instead assume that ϵ follows a logistic distribution, then $\Pr(\epsilon \le t) = \frac{e^t}{1+e^t}$, so

$$\Pr(Y_i \le k \mid X_i = x) = \Pr(\epsilon \le \alpha_k - x^\top \beta) = \frac{e^{\alpha_k - x^\top \beta}}{1 + e^{\alpha_k - x^\top \beta}}.$$

Likelihood

Since Y_i is discrete, $\Pr(Y_i = k) = \Pr(Z_i \le \alpha_k) - \Pr(Z_i \le \alpha_{k-1})$. The likelihood is therefore:

$$\begin{aligned} L(\beta) &= \prod_{i=1}^{N} \Pr(Y_i = y_i \mid X_i = x) \\ &= \prod_i \left[\Pr(Z_i \le y_i \mid X_i = x) - \Pr(Z_i \le y_i - 1 \mid X_i = x)\right] \\ &= \prod_i \left[\Phi(\alpha_{y_i} - x^\top \beta) - \Phi(\alpha_{y_i - 1} - x^\top \beta)\right] \end{aligned}$$

with the convention $\alpha_0 = -\infty$ and $\alpha_K = +\infty$ so that $\Phi(\alpha_0 - x^\top \beta) = 0$ and $\Phi(\alpha_K - x^\top \beta) = 1$. For K categories, we only need $K-1$ threshold parameters $\alpha_1 < \cdots < \alpha_{K-1}$; α_0 and α_K do not exist as free parameters but are boundary conventions to ensure probabilities are well-defined. For example, with $K = 4$: $\Pr(Y_i = 1) = \Phi(\alpha_1 - x^\top \beta)$ (since $\Phi(\alpha_0 - x^\top \beta) = 0$), and $\sum_{k=1}^{4} \Pr(Y_i = k) = \Phi(\alpha_4 - x^\top \beta) = 1$ forces $\alpha_4 = +\infty$.

Sign Convention and Software

Somewhat counter-intuitively, the sign of $x^\top \beta$ is *negative* in the expression $\Phi(\Pr(Y_i \le k \mid x)) = \alpha_k - x^\top \beta$. This ensures that increasing $x^\top \beta$ tends to *increase* the category, since the probability being modeled is $\Pr(Y_i \le k)$ — the probability of being in a *smaller* category. However, different software packages may disagree on the sign. For example, `VGAM` in R uses the convention $\text{logit}(\Pr(Y_i \le k \mid x)) = \alpha_k + x^\top \beta$, so that increasing the linear predictor tends to *increase* the probability of lying in the lower class. Always check the `Names of linear predictors` heading in the output to confirm which convention is being used.

Advantages and Disadvantages

Advantages. The cumulative link model has several advantages over the multinomial logit: (1) it naturally respects the ordering of categories through cumulative probabilities; (2) it has substantially fewer parameters, as the same β is shared across all categories and only the intercepts α_k

vary; (3) the *proportional odds property* gives a clean interpretation — the odds ratio comparing $\Pr(Y_i \le k)$ between two covariate vectors x and x' is the same for all k:

$$\frac{\mathrm{Odds}(Y_i \le k \mid x)}{\mathrm{Odds}(Y_i \le k \mid x')} = e^{(x-x')^\top \beta}.$$

Disadvantages: Varying Slopes. If we relax the proportional odds assumption and allow β to vary by threshold, i.e., $\Pr(Y_i \le k \mid X_i = x) = \Phi(\alpha_k - x^\top \beta_k)$, three serious problems arise:

- **Monotonicity can be violated.** The cumulative probabilities must satisfy $\Pr(Y_i \le 1) \le \Pr(Y_i \le 2) \le \cdots \le \Pr(Y_i \le K-1)$ for all x. With shared β, this reduces to $\alpha_k - x^\top\beta > \alpha_{k-1} - x^\top\beta \iff \alpha_k > \alpha_{k-1}$, which always holds by construction. However, with threshold-specific β_k, we need $\alpha_k - x^\top\beta_k \ge \alpha_{k-1} - x^\top\beta_{k-1}$ for *all* x, a complex constraint that depends on the covariate values. For a given set of estimated $\{\beta_k\}$, there may exist covariate values x for which $\Phi(\alpha_k - x^\top\beta_k) < \Phi(\alpha_{k-1} - x^\top\beta_{k-1})$, implying $\Pr(Y_i = k) = \Pr(Y_i \le k) - \Pr(Y_i \le k-1) < 0$. This is a fundamental problem: the model can produce invalid probability assignments.
- **Parameter count explodes.** The model goes from $P + (K-1)$ parameters to $P(K-1) + (K-1) = (K-1)(P+1)$, the same as the multinomial logit, eliminating the parsimony advantage. If we are willing to estimate that many parameters, we might as well use the multinomial logit and abandon the ordinal structure entirely.
- **Interpretability is lost.** There is no longer a single coefficient β_j summarizing the effect of predictor j across all thresholds; the proportional odds property no longer holds.

For these reasons, the proportional odds assumption (constant β across thresholds) is strongly preferred in practice, and should be tested rather than abandoned without justification.

Continuation Ratio Models

Motivation While cumulative link models are convenient, they are somewhat rigid in that they assume that the way the predictors separate (say) the first and second categories is the same as the way they separate the $(K-1)$th and Kth. This would not be reasonable, for example, if "having a lot of money" stops people from being extremely depressed, but is insufficient to make someone never depressed. In theory this would be handled by having a separate β_k for the different categories, but as we have seen this can run into serious issues.

Continuation ratio models take a different approach by modeling the probability of advancing to higher categories, given that an observation has reached at least the current category. Mathematically, what we are modeling is:

$$\lambda_k(x) = \Pr(Y_i = k \mid Y_i \ge k,\ X_i = x)$$

which represents the probability of "stopping" at category k, given that we have reached that category.

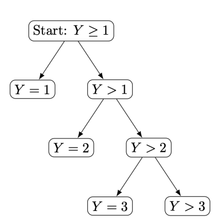

Figure 3: Schematic showing the basic idea of a continuation ratio model. At each level, we go left with probability $\lambda_k(x)$ and right with probability $1 - \lambda_k(x)$.

Stick-Breaking Construction

The category probabilities $\pi_k(x) = \Pr(Y_i = k \mid X_i = x)$ can be recovered from the conditional stopping probabilities via the *stick-breaking construction*:

$$\pi_k(x) = \lambda_k(x) \prod_{j=1}^{k-1} \{1 - \lambda_j(x)\}.$$

To see this, note that

$$\pi_k(x) = \Pr(Y_i = k \mid Y_i \geq k)\ \Pr(Y_i \geq k) = \lambda_k(x)\ \Pr(Y_i \geq k).$$

Expanding $\Pr(Y_i \geq k)$ recursively:

$$\Pr(Y_i \geq k) = \Pr(Y_i \geq k \mid Y_i \geq k-1)\ \Pr(Y_i \geq k-1) = (1 - \lambda_{k-1}(x))\ \Pr(Y_i \geq k-1).$$

Repeating this yields

$$\Pr(Y_i \geq k) = \prod_{j=1}^{k-1} (1 - \lambda_j(x)) \cdot \underbrace{\Pr(Y_i \geq 1)}_{=1} = \prod_{j=1}^{k-1} \{1 - \lambda_j(x)\}.$$

Substituting back gives the result.

Link Functions

This sequential perspective makes them particularly suitable for processes where categories represent progressive stages or levels. The model typically uses either the logit link:

$$\operatorname{logit}(\lambda_k(x)) = \alpha_k - x^\top \beta_k$$

or alternatively, using the complementary log-log link:

$$\log(-\log(1 - \lambda_k(x))) = \alpha_k - x^\top \beta_k.$$

Note that we can keep the coefficient vector β_k constant across categories ($\beta_k = \beta$ for all β, as a sort of parallel-trends assumption) or we can estimate a separate β_k for each category. Unlike with the cumulative link models, there is no risk of anything going wrong here, as there are no constraints on the parameter space.

Likelihood and Estimation

Another practical advantage of the continuation ratio model is that they are very easy to fit. The likelihood function is given by

$$\begin{aligned}
L(\beta, \alpha) &= \prod_{i=1}^{N} \left[\prod_{k=1}^{Y_i - 1} (1 - \lambda_k(x_i)) \right] \times \lambda_{Y_i}(x_i) \\
&= \prod_{i=1}^{N} \left[\prod_{k=1}^{Y_i - 1} \left(1 - \frac{e^{\alpha_k - x_i^\top \beta}}{1 + e^{\alpha_k - x_i^\top \beta}} \right) \right] \times \frac{e^{\alpha_{Y_i} - x_i^\top \beta}}{1 + e^{\alpha_{Y_i} - x_i^\top \beta}} \\
&= \prod_{k=1}^{K-1} \prod_{i:\, Y_i \geq k} \left(\frac{e^{\alpha_k - x_i^\top \beta}}{1 + e^{\alpha_k - x_i^\top \beta}} \right)^{I(Y_i = k)} \left(\frac{1}{1 + e^{\alpha_k - x_i^\top \beta}} \right)^{I(Y_i > k)}
\end{aligned}$$

good luck :)

Note that the outer product runs to $K-1$ rather than K: at $k = K$ (the highest category), $I(Y_i > K)$ is never satisfied, so that factor contributes nothing. This represents a sequence of $K-1$ separate logistic regressions, and can therefore be fit by converting the ordinal observations Y_i into a sequence of Y_i-many binary variables $(Z_{ik} : k \leq Y_i)$ where $Z_{ik} = 1$ for $Y_i = k$ and $Z_{ik} = 0$ if $Y_i > k$. The model for these binary variables is simply a logistic regression, and can therefore be fit using any software that permits fitting logistic regressions (or probit, cloglog, cauchit).

Non-Parallel (β_k unrestricted). When each category has its own coefficient vector β_k, the model for each layer is:

$$\text{logit}(\Pr(Z_{ik} = 1)) = \alpha_k - x_i^\top \beta_k, \qquad k = 1, \dots, K-1,$$

fit separately on $\{i : Y_i \geq k\}$. The likelihood factorizes completely:

$$L(\beta_1, \dots, \beta_{K-1}, \alpha) = \prod_{k=1}^{K-1} \underbrace{\prod_{i:\, Y_i \geq k} \left(\frac{e^{\alpha_k - x_i^\top \beta_k}}{1 + e^{\alpha_k - x_i^\top \beta_k}} \right)^{I(Y_i = k)} \left(\frac{1}{1 + e^{\alpha_k - x_i^\top \beta_k}} \right)^{I(Y_i > k)}}_{L_k(\alpha_k, \beta_k)}$$

Each L_k depends only on (α_k, β_k) and is independent of the other layers, so we can maximize each one separately: for each k, fit a standard logistic regression of $Z_{ik} = I(Y_i = k)$ on x_i using only the observations with $Y_i \geq k$.

Why Each Layer Is a Logistic Regression Fix a layer k and examine what L_k looks like. The set of individuals who participate in this layer is $\{i : Y_i \geq k\}$—this is the *risk set* for layer k. Within this group, define a new binary variable:

$$Z_{ik} = \begin{cases} 1 & \text{if } Y_i = k \quad \text{(stopped)}, \\ 0 & \text{if } Y_i > k \quad \text{(continued)}. \end{cases}$$

Then L_k can be rewritten as

$$\begin{aligned} L_k &= \prod_{i:\, Y_i \geq k} (\frac{e^{\alpha_k - x_i^\top \beta_k}}{1 + e^{\alpha_k - x_i^\top \beta_k}})^{Z_{ik}} (\frac{1}{1 + e^{\alpha_k - x_i^\top \beta_k}})^{1 - Z_{ik}} \\ &= \prod_{i:\, Y_i \geq k} \Pr(Z_{ik} = 1)^{Z_{ik}} \big(1 - \Pr(Z_{ik} = 1)\big)^{1 - Z_{ik}}. \end{aligned}$$

This is exactly a standard Bernoulli likelihood with success probability $\text{logit}\big(\Pr(Z_{ik} = 1)\big) = \alpha_k - x_i^\top \beta_k$. In other words, at layer k, Z_{ik} follows a logistic regression on x_i with intercept α_k and coefficient vector $-\beta_k$ (or $-\beta$ in the parallel model).

Parallel ($\beta_k = \beta$ for all k). When all categories share the same β, the model is:

$$\text{logit}(\Pr(Z_{ik} = 1)) = \sum_{j=1}^{K-1} \alpha_j D_j - x_i^\top \beta$$

where $D_j = I(k = j)$ are dummy variables for the layer index. The likelihood is:

$$\begin{aligned} L(\beta, \alpha_1, \dots, \alpha_{K-1}) &= \prod_{k=1}^{K-1} \prod_{i:\, Y_i \geq k} (\frac{e^{\alpha_k - x_i^\top \beta}}{1 + e^{\alpha_k - x_i^\top \beta}})^{I(Y_i = k)} (\frac{1}{1 + e^{\alpha_k - x_i^\top \beta}})^{I(Y_i > k)} \\ &= \prod_{k=1}^{K-1} \prod_{i:\, Y_i \geq k} \Pr(Z_{ik} = 1)^{Z_{ik}} \big(1 - \Pr(Z_{ik} = 1)\big)^{1 - Z_{ik}} \end{aligned}$$

good luck :)

The factors L_k are coupled through the shared β and cannot be optimized separately. Instead, we stack all $K-1$ sub-datasets into a single dataset with response Z_{ik} and fit a single logistic regression: β is shared across all layers, while different intercepts α_k are achieved through the dummy variables.

Proportional Hazards Equivalence

A continuation ratio model with $\beta_k = \beta$ fixed across k and using the complementary log-log link has been called the *proportional hazards model* for ordinal data. This model is exactly equivalent to a cumulative link model with the complementary log-log link.
The CR model with cloglog link (parallel) specifies:

$$\log(-\log(1-\lambda_k(x))) = \alpha_k - x^\top \beta$$

where $\lambda_k(x) = \Pr(Y_i = k \mid Y_i \geq k, x)$, so that:

$$1 - \lambda_k(x) = \Pr(Y_i > k \mid Y_i \geq k, x) = \exp(-\exp(\alpha_k - x^\top \beta)).$$

By the stick-breaking construction, the survival probability past category k is:

$$\Pr(Y_i > k \mid x) = \prod_{j=1}^{k}(1-\lambda_j(x)) = \prod_{j=1}^{k} \exp(-\exp(\alpha_j - x^\top \beta)) = \exp\left(-\sum_{j=1}^{k} e^{\alpha_j - x^\top \beta}\right) = \exp\left(-e^{-x^\top \beta} \sum_{j=1}^{k} e^{\alpha_j}\right).$$

Therefore:

$$\Pr(Y_i \leq k \mid x) = 1 - \exp\left(-e^{-x^\top \beta} \sum_{j=1}^{k} e^{\alpha_j}\right).$$

Applying the complementary log-log transformation to both sides:

$$\log(-\log(1 - \Pr(Y_i \leq k \mid x))) = \log\left(e^{-x^\top \beta} \sum_{j=1}^{k} e^{\alpha_j}\right) = \log\left(\sum_{j=1}^{k} e^{\alpha_j}\right) - x^\top \beta.$$

Setting $\tilde{\alpha}_k = \log\left(\sum_{j=1}^{k} e^{\alpha_j}\right)$, we obtain:

$$\log(-\log(1 - \Pr(Y_i \leq k \mid x))) = \tilde{\alpha}_k - x^\top \beta.$$

This is exactly the **cumulative link model with the complementary log-log link**, with the same coefficient vector β and reparameterized intercepts $\tilde{\alpha}_k$. Since $\sum_{j=1}^{k} e^{\alpha_j}$ is strictly increasing in k, we have $\tilde{\alpha}_1 < \tilde{\alpha}_2 < \cdots < \tilde{\alpha}_{K-1}$, so the threshold ordering constraint of the cumulative link model is automatically satisfied.
Remark. From Lecture 9, under proportional hazards, $S(t \mid x) = S_0(t)^{\exp(x^\top \beta)}$ where $S_0(t) = \exp\bigl(-\int_0^t h_0(z)\,dz\bigr)$ is the baseline survival function. Equivalently, this gives a log-log link:

$$\log\bigl\{-\log S(t \mid x)\bigr\} = \log\bigl\{-\log S_0(t)\bigr\} + x^\top \beta.$$

Now, let's consider the relationship with

$$\log(-\log(1-\lambda_k(x))) = \alpha_k - x^\top \beta.$$

From the above equation, we have

$$1 - \lambda_k(x) = e^{-e^{\alpha_k - x^T \beta}}.$$

Also, under a survival background,

$$1 - \lambda_k(x) = P(T > k | T \geq k, x) = \frac{P(T > k \mid x)}{P(T \geq k \mid x)} = \frac{P(T > k \mid x)}{P(T > k-1 \mid x)} = \frac{S(k|x)}{S(k-1|x)}.$$

We know the survival function satisfies:

$$S(k \mid x) = (S_0(k))^{\exp(-x^\top \beta)}.$$

Finally, we have

$$1 - \lambda_k(x) = \frac{S(k|x)}{S(k-1|x)} = \left(\frac{S_0(k)}{S_0(k-1)} \right)^{\exp(-x^\top \beta)}.$$

If we let $\exp(-\exp(\alpha_k)) = \frac{S_0(k)}{S_0(k-1)}$, the two equations are equal.
Additional, $1 - \lambda_k(x) = \frac{S(k|x)}{S(k-1|x)}$ can be proved by the following idea: we know

$$S(k \mid x) = P(T > k|x) = \prod_{j=1}^{k} (1 - \lambda_j(x)) = S(k-1|x)(1 - \lambda_k),$$

based on the above equation $\Pr(Y_i > k \mid x) = \prod_{j=1}^{k} (1 - \lambda_j(x))$.

good luck :)

7 Moment-Based Estimator

Semiparametric Methods

Up to this point, we have mainly been using parametric models:

$$L(\theta) = \prod_{i=1}^{N} f_\theta(Y_i \mid X_i)$$

and we have primarily been estimating θ by maximizing the likelihood.
In semiparametric statistics, we instead consider a family of distributions for the data that is of the form

$$\{G_{\theta,\eta} : \theta \in \Theta,\, \eta \in H\}$$

where η is *infinite dimensional*. We have two strategies:

1. Try to estimate η as part of the estimation process for θ.
2. Try to bypass estimating η in the first place, and try to just directly estimate the θ that we care about.

Weighted Least Squares

Consider a linear regression model where

$$Y_i = X_i^\top \beta + \varepsilon_i.$$

We assume that $\mathbb{E}[\varepsilon_i \mid X_i] = 0$ and $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2(X_i)$ with, say, $\log \sigma^2(x) = x^\top \gamma$.
In this case:

- The parameter of interest is $\theta = \beta$, the regression coefficients.
- A nuisance parameter is γ, the conditional variance parameter.
- Another nuisance parameter is the distribution of $R_i = (Y_i - X_i^\top \beta)/\sigma(X_i)$ conditional on X_i. This is an arbitrary conditional distribution $f(r \mid x)$ subject to $\int r\, f(r \mid x)\, dr = 0$ and $\int r^2\, f(r \mid x)\, dr = 1$.

We could estimate β using ordinary least squares, which gives a consistent estimator even when the variance function is misspecified. Alternatively, if efficiency is a concern, we could:

1. Initialize $\hat\gamma$ somehow (setting all coefficients to zero works).
2. Use weighted least squares with weights $w_i = 1/\hat\sigma^2(X_i)$ to estimate β.
3. Do a linear regression on $V_i = \log(Y_i - X_i^\top \hat\beta)^2 = X_i^\top \gamma + \nu_i$ to update $\hat\gamma$.
4. Repeat steps 2–3 many times.

Note that this involves (i) a parameter of interest β; (ii) a nuisance parameter γ that we would like to estimate to improve inference on β; and (iii) a nuisance parameter $f(r \mid x)$ that we do not try to estimate.

Remark. WLS requires knowledge of $\sigma^2(X_i)$ to construct weights, but in practice we do not know it. The solution is an iterative algorithm that estimates β and γ (the variance parameter) simultaneously.

1. **Step 1:** Initialize $\hat{\gamma} = 0$. Since the model assumes $\log \sigma^2(x) = x^\top \gamma$, setting $\hat{\gamma} = 0$ gives $\hat{\sigma}^2(x) = \exp(0) = 1$, i.e., homoskedasticity. The WLS weights become $w_i = 1/\hat{\sigma}^2(X_i) = 1$, reducing to OLS.

2. **Step 2:** Using the current $\hat{\sigma}^2(X_i)$, perform WLS to obtain $\hat{\beta}$. In the first iteration this is simply OLS.

3. **Step 3:** Given $\hat{\beta}$, compute residuals $\hat{\varepsilon}_i = Y_i - X_i^\top \hat{\beta}$. We then regress $V_i = \log(\hat{\varepsilon}_i^2)$ on X_i to update $\hat{\gamma}$. The motivation is that the true relationship is $\varepsilon_i = \sigma(X_i) \cdot R_i$, so $\varepsilon_i^2 = \sigma^2(X_i) \cdot R_i^2$. Taking logs:
$$\log \varepsilon_i^2 = \log \sigma^2(X_i) + \log R_i^2 = X_i^\top \gamma + \nu_i$$
where $\nu_i = \log R_i^2$ is an error term. This is a linear regression with coefficients γ, so regressing V_i on X_i by OLS updates $\hat{\gamma}$.

4. **Step 4:** Compute the updated $\hat{\sigma}^2(X_i) = \exp(X_i^\top \hat{\gamma})$ and return to Step 2. Repeat until convergence.

This is essentially an **alternating optimization**: fix γ to estimate β, then fix β to estimate γ, iterating until convergence. A key insight from semiparametric theory is that $\hat{\gamma}$ does not need to be estimated very precisely; as long as it is $\sqrt{N}$-consistent, the asymptotic efficiency of $\hat{\beta}$ is the same as if the true γ were known. That is, the estimation error in the nuisance parameter "disappears for free" at the first-order asymptotic level.

Remark. Using OLS when the true data is heteroskedastic (i.e., $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2(X_i)$, where WLS should be used) but we ignore the heteroskedastic structure:

- *Consistency*: OLS only relies on $\mathbb{E}[\varepsilon_i \mid X_i] = 0$ and requires no variance information, so even if the data is heteroskedastic, OLS remains consistent.
- *Efficiency*: The price is that OLS is not efficient. The true asymptotic variance is the sandwich form
$$\mathbb{E}[X_i X_i^\top]^{-1} \mathbb{E}[X_i X_i^\top \sigma^2(X_i)] \mathbb{E}[X_i X_i^\top]^{-1},$$
not the homoskedastic formula $\sigma^2 \mathbb{E}[X_i X_i^\top]^{-1}$.

The takeaway is: regardless of the weights used (including all weights equal to 1, i.e., OLS), as long as the mean is correctly specified, $\hat{\beta}$ is consistent — only the efficiency differs. This is the power of the M-estimation framework: all such cases are unified under the sandwich form.

Remark. *Why do observations with large variance receive smaller weights in WLS?* This is a signal-to-noise ratio argument. In $Y_i = X_i^\top \beta + \varepsilon_i$, the information each observation provides about β depends on how large the noise ε_i is. If $\sigma^2(X_i)$ is large, Y_i fluctuates widely and it is difficult to extract information about β — the signal is drowned out by noise. Conversely, if $\sigma^2(X_i)$ is small, Y_i clusters tightly around $X_i^\top \beta$, providing precise information about β.
Formally, WLS minimizes $\sum (Y_i - X_i^\top \beta)^2 / \sigma^2(X_i)$, which standardizes each residual: $(Y_i - X_i^\top \beta)/\sigma(X_i)$ has unit variance at the true β regardless of X_i. After standardization, all observations contribute "fairly" to the objective, and no single observation dominates the optimization simply because it has large natural variance. OLS does not perform this standardization, so observations with large variance produce large residuals that disproportionately influence the objective — exactly the opposite of what we want.

Remark. *Why assume* $\log \sigma^2(x) = x^\top \gamma$*?* This is not a requirement but a modeling choice, motivated by three considerations:

- *Positivity*: The variance $\sigma^2(x)$ must be positive. If we directly assumed $\sigma^2(x) = x^\top\gamma$, we could obtain negative values, which are physically meaningless. After taking logs, $\log\sigma^2$ can be any real number, and $\sigma^2(x) = \exp(x^\top\gamma)$ is automatically positive. This is the same idea as using a log link in Poisson regression to ensure a positive mean.
- *Empirical regularity*: In many real datasets, the variance grows exponentially rather than linearly with the mean or covariates. For example, in income data, the variance of high-income groups is much larger than that of low-income groups, and the gap is often multiplicative. The model $\log\sigma^2 = x^\top\gamma$ captures this multiplicative structure.
- *Computational convenience*: This assumption turns the Step 3 estimation of γ into a simple linear regression: $\log\hat{\varepsilon}_i^2 = X_i^\top\gamma + \nu_i$, solvable directly by OLS. Other parameterizations for γ might require nonlinear optimization, making the algorithm considerably more complex.

However, this is just one specific example. One could use alternative parameterizations, such as $\sigma^2(x) = \exp(h(x)^\top\gamma)$ where $h(x)$ is some basis function transformation, or even nonparametric methods to estimate the variance function. The core idea remains the same: as long as one obtains a $\sqrt{N}$-consistent estimate $\hat{\sigma}^2(x)$, the resulting WLS estimator $\hat{\beta}$ will be asymptotically efficient.

OLS Consistency Under Variance Misspecification

Consider the model

$$Y_i = X_i^\top\beta_0 + \varepsilon_i.$$

The only key assumption is $\mathbb{E}\big[\varepsilon_i \mid X_i\big] = 0$. The variance $\mathrm{Var}(\varepsilon_i \mid X_i) = \sigma^2(X_i)$ is unknown and may depend on X_i (i.e., heteroskedasticity), but we **do not need** any assumptions on it.

Proof of Consistency. OLS solves the normal equation $\sum X_i(Y_i - X_i^\top\beta) = 0$, with closed-form solution $\hat{\beta} = (X^\top X)^{-1}X^\top Y$. Substituting $Y_i = X_i^\top\beta_0 + \varepsilon_i$:

$$\hat{\beta} - \beta_0 = \big(\frac{1}{N}\sum X_iX_i^\top\big)^{-1}\cdot\big(\frac{1}{N}\sum X_i\varepsilon_i\big).$$

First term: by the LLN, $\frac{1}{N}\sum X_iX_i^\top \xrightarrow{p} \mathbb{E}\big[X_iX_i^\top\big]$ (assumed invertible). Second term: by the LLN, $\frac{1}{N}\sum X_i\varepsilon_i \xrightarrow{p} \mathbb{E}\big[X_i\varepsilon_i\big]$. By iterated expectation:

$$\mathbb{E}\big[X_i\varepsilon_i\big] = \mathbb{E}\big[X_i\cdot\mathbb{E}[\varepsilon_i \mid X_i]\big] = \mathbb{E}\big[X_i\cdot 0\big] = 0.$$

Therefore $\hat{\beta} - \beta_0 \xrightarrow{p} \mathbb{E}\big[X_iX_i^\top\big]^{-1}\cdot 0 = 0$, i.e., $\hat{\beta} \xrightarrow{p} \beta_0$.

Remark. The entire derivation makes no use of the variance. Regardless of whether $\sigma^2(X_i)$ is a constant, a function of X_i, or arbitrary, consistency is unaffected.

True Asymptotic Variance (Sandwich Form). Starting from the same decomposition, multiply by $\sqrt{N}$:

$$\sqrt{N}(\hat{\beta} - \beta_0) = \big(\frac{1}{N}\sum X_iX_i^\top\big)^{-1}\cdot\big(\frac{1}{\sqrt{N}}\sum X_i\varepsilon_i\big).$$

First term: $\frac{1}{N}\sum X_iX_i^\top \xrightarrow{p} \mathbb{E}\big[X_iX_i^\top\big]^{-1}$ (as above). Second term: by the CLT, $\frac{1}{\sqrt{N}}\sum X_i\varepsilon_i \xrightarrow{d} N\big(0,\, \mathrm{Var}(X_i\varepsilon_i)\big)$. Since the mean is zero, the variance equals the second moment:

$$\mathrm{Var}(X_i\varepsilon_i) = \mathbb{E}\big[X_i\varepsilon_i^2X_i^\top\big].$$

By iterated expectation:

$$\mathbb{E}\big[X_i\varepsilon_i^2X_i^\top\big] = \mathbb{E}\big[X_iX_i^\top\cdot\mathbb{E}[\varepsilon_i^2 \mid X_i]\big] = \mathbb{E}\big[X_iX_i^\top\sigma^2(X_i)\big].$$

By Slutsky's theorem, combining both terms:

$$\sqrt{N}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, V)$$

where

$$\boxed{V = \mathbb{E}\big[X_i X_i^\top\big]^{-1} \cdot \mathbb{E}\big[X_i X_i^\top \sigma^2(X_i)\big] \cdot \mathbb{E}\big[X_i X_i^\top\big]^{-1}}$$

This is the **sandwich form**: $A^{-1}BA^{-\top}$, where $A = \mathbb{E}\big[X_i X_i^\top\big]$ and $B = \mathbb{E}\big[X_i X_i^\top \sigma^2(X_i)\big]$.
Homoskedastic Special Case. If $\sigma^2(X_i) = \sigma^2$ (constant), then $B = \sigma^2\mathbb{E}\big[X_i X_i^\top\big]$, and the sandwich simplifies to

$$V = \mathbb{E}\big[X_i X_i^\top\big]^{-1} \cdot \sigma^2\mathbb{E}\big[X_i X_i^\top\big] \cdot \mathbb{E}\big[X_i X_i^\top\big]^{-1} = \sigma^2\mathbb{E}\big[X_i X_i^\top\big]^{-1}.$$

This is the classical OLS asymptotic variance. Under heteroskedasticity, $\sigma^2(X_i)$ cannot be cancelled, so the full sandwich form is required.

Moment Based Estimators

A standard way to construct semiparametric estimators is to identify the parameters of interest as *solutions to a set of moment equations*. For example, suppose we have a linear model

$$Y_i = X_i^\top \beta_0 + \epsilon_i.$$

Then, evidently, $\mathbb{E}\big[Y_i - X_i^\top \beta_0\big] = 0$; that is, based on this equation we know that β_0 must lie in the set

$$\big\{\beta : \mathbb{E}\big[Y_i - X_i^\top \beta\big] = 0\big\}.$$

Of course, if $\beta \in \mathbb{R}^P$, then having just one equation is not sufficient to pin down β_0. More generally, we might notice that for **any** function $A(x) : \mathbb{R}^P \to \mathbb{R}^P$ we can write

$$\mathbb{E}\big[A(X_i)(Y_i - X_i^\top \beta)\big] = \mathbb{E}\big[A(X_i)\mathbb{E}\big[Y_i - X_i^\top \beta \mid X_i\big]\big] = \mathbf{0}.$$

With P equations and P unknowns, this should be enough to pin down β. To get an actual estimator out of this, we simply approximate the expectation with respect to the population distribution using the sampling distribution, and define $\hat{\beta}$ as the solution to

$$\frac{1}{N}\sum_{i=1}^{N} A(X_i)(Y_i - X_i^\top \beta) = \mathbf{0}.$$

Note that we actually have a surprising amount of degrees of freedom for how we choose the mapping $A(X_i)$! As a special case, setting $A(X_i) = X_i$ recovers ordinary least squares. For homoskedastic settings, $A(X_i) = X_i$ is the optimal choice, but in other settings we might try to choose $A(x)$ in a way that minimizes the asymptotic variance of the resulting $\hat{\beta}$.
Remark. If we set the first component of $A(X_i)$ to be 1, i.e., $A(X_i) = (1, \ldots)^\top$, then the first row of the system $\mathbb{E}\big[A(X_i)(Y_i - X_i^\top \beta)\big] = \mathbf{0}$ reduces to $\mathbb{E}\big[Y_i - X_i^\top \beta\big] = 0$, so the single moment equation is automatically included as part of the P equations.

M-estimators

The technique of M-estimation provides a unifying framework for many common semiparametric estimators, including maximum likelihood estimators, moment-based estimators, and robust estimators. The "M" stands for either "maximum" or "minimum," as they are derived by maximizing/minimizing objective functions.

Definition. An M-estimator $\hat{\theta}$ is defined as the value of θ that minimizes an arbitrarily-defined objective function:

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \frac{1}{N} \sum_{i=1}^{N} \rho(Y_i, X_i;\, \theta).$$

We should expect $\hat{\theta}$ to be a reasonable estimator of the true θ_0 as long as θ_0 satisfies the population-level version:

$$\theta_0 = \arg\min_{\theta \in \Theta} \mathbb{E}_{\theta_0}\big[\rho(Y_i, X_i;\, \theta)\big].$$

Usually (although not always) the problem is regular enough that we can equivalently write: $\hat{\theta}$ solves

$$\frac{1}{N} \sum_{i=1}^{N} \psi(Y_i, X_i;\, \theta) = 0$$

where $\psi(y, x;\, \theta) = \frac{\partial}{\partial \theta} \rho(y, x;\, \theta)$. Estimators that solve estimating equations are occasionally called Z-estimators (the "Z" standing for "zero"), but we will use the term M-estimator for either.

Examples

Show that the following estimators are examples of M-estimators, and identify the respective $\psi(y, x;\, \theta)$ for each:

MLE. The maximum likelihood estimator $\hat{\theta}$ under some parametric model.

Sample Quantile. The sample 75th percentile of the (marginal) distribution of the Y_i's (you can assume for simplicity that $N = 100$).

OLS. OLS minimizes the sum of squared residuals, so

$$\rho(y, x;\, \beta) = \frac{(y - x^\top \beta)^2}{2}.$$

The division by 2 is for convenience when differentiating and does not affect the minimizer. Differentiating with respect to β by the chain rule:

$$\psi(y, x;\, \beta) = \frac{\partial \rho}{\partial \beta} = (y - x^\top \beta) \cdot (-x) = -x(y - x^\top \beta).$$

Setting $\frac{1}{N} \sum \psi = 0$ and dropping the irrelevant negative sign, we recover the OLS normal equation:

$$\frac{1}{N} \sum X_i (Y_i - X_i^\top \beta) = 0.$$

So we may write $\psi(y, x;\, \beta) = x(y - x^\top \beta)$.

WLS. WLS minimizes the weighted sum of squared residuals, where observations with large variance receive smaller weight:

$$\rho(y, x;\, \beta) = \frac{(y - x^\top \beta)^2}{2\sigma^2(x)}.$$

Note that $\sigma^2(x)$ is a function of x only and does not depend on β, so it acts as a constant when differentiating:

$$\psi(y, x;\, \beta) = \frac{\partial \rho}{\partial \beta} = \frac{-x(y - x^\top \beta)}{\sigma^2(x)}.$$

Dropping the negative sign, the estimating equation becomes

$$\frac{1}{N} \sum \frac{X_i(Y_i - X_i^\top \beta)}{\sigma^2(X_i)} = 0.$$

Compared to OLS, the only difference is that each observation is weighted by $1/\sigma^2(X_i)$.

Asymptotic Normality of M-Estimators

A convenient property of M-estimators is that they are (usually) asymptotically normal, under very minimal assumptions. The argument for this mirrors the asymptotic normality of the MLE. Assume that $\psi(y, x;\, \theta)$ has a continuous Jacobian matrix $\dot{\psi}(y, x;\, \theta) = \frac{\partial}{\partial \theta}\psi(y, x;\, \theta)$ (plus whatever regularity conditions are required to make the various arguments go through) and define

$$A(\theta) = \mathbb{E}\big[\dot{\psi}(Y_i, X_i;\, \theta)\big] \qquad \text{and} \qquad B(\theta) = \mathbb{E}\big[\psi(Y_i, X_i;\, \theta)\, \psi(Y_i, X_i;\, \theta)^\top\big].$$

Show that

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \text{Normal}(0,\, V)$$

where $V = A(\theta_0)^{-1} B(\theta_0)\, A(\theta_0)^{-\top}$.
The asymptotic variance $V = A(\theta_0)^{-1} B(\theta_0)\, A(\theta_0)^{-T}$ is often called the "sandwich" form due to its structure. There are several ways to estimate the sandwich matrix, but an obvious choice is

$$\hat{V} = \hat{A}^{-1} \hat{B}\, \hat{A}^{-\top}$$

where

$$\hat{A} = \frac{1}{N} \sum_{i=1}^{N} \dot{\psi}(Y_i, X_i;\, \hat{\theta}) \qquad \text{and} \qquad \hat{B} = \frac{1}{N} \sum_{i=1}^{N} \psi(Y_i, X_i;\, \hat{\theta})\, \psi(Y_i, X_i;\, \hat{\theta})^\top.$$

Maximum Likelihood as M-Estimation

We now explore ML estimation from the perspective of M-estimation, particularly focusing on the asymptotic properties of the MLE under mis-specification. Assume that $(X_i, Y_i) \stackrel{\text{iid}}{\sim} F_0$.
MLE as M-Estimator. The MLE can be viewed as an M-estimator with $\psi(y, x;\, \theta) = \frac{\partial}{\partial \theta} \log f_\theta(y \mid x)$, where $f_\theta(y \mid x)$ is the conditional density function specified by our model; equivalently, we can take $\rho(y, x;\, \theta) = -\log f_\theta(y \mid x)$.
KL Projection. Without any assumptions about model correctness, the MLE is an estimator of the population-level quantity:

$$\theta^\star = \arg\min_\theta - \int \log f_\theta(y \mid x)\; F_0(dx,\, dy).$$

The distribution $F_{\theta^\star}$ is sometimes called the *Kullback–Leibler projection* of F_0 onto $\big\{F_\theta : \theta \in \Theta\big\}$.
Sandwich Matrix Under Misspecification. The matrices $A(\theta^\star)$ and $B(\theta^\star)$ for the MLE as an M-estimator under misspecification are:

$$A(\theta^\star) = \mathbb{E}\big[\frac{\partial^2}{\partial \theta\, \partial \theta^\top} \log f_{\theta^\star}(Y_i \mid X_i)\big]$$

and

$$B(\theta^\star) = \mathbb{E}\big[\frac{\partial}{\partial\theta}\log f_{\theta^\star}(Y \mid X)\ \frac{\partial}{\partial\theta^\top}\log f_{\theta^\star}(Y \mid X)\big].$$

Correct Specification. Under correct specification, the sandwich matrix simplifies to the usual inverse Fisher information. This is because the Bartlett identity gives $A(\theta_0) = -B(\theta_0) = -\mathcal{I}(\theta_0)$, so

$$V = A(\theta_0)^{-1}B(\theta_0)\,A(\theta_0)^{-\top} = \mathcal{I}(\theta_0)^{-1}.$$

Remark. A practical question is whether to default to using the sandwich estimator $\hat{V} = \hat{A}^{-1}\hat{B}\,\hat{A}^{-\top}$ as the asymptotic covariance for $\hat{\theta}$, as opposed to the observed Fisher information $\mathcal{J}(\hat{\theta})^{-1}$. The sandwich estimator is *robust*: it remains valid even under misspecification, whereas the Fisher information is only correct when the model is correctly specified. However, the sandwich estimator typically has higher variance in finite samples, since it involves estimating two matrices instead of one. In practice, the choice reflects a bias-variance trade-off: the sandwich is safer but noisier; the Fisher information is more precise but can be badly wrong if the model is misspecified.

The Nonparametric Bootstrap

As an alternative to estimating the sandwich matrix directly, the nonparametric bootstrap can be used to get a valid estimate of the variance of $\hat{\beta}$. The nonparametric bootstrap estimate of the variance will, in fact, be asymptotically equivalent to using the sandwich matrix. So, if you have CPU cycles to burn, you can get generally valid inference, under only the assumption $\mathbb{E}\big[Y_i \mid X_i\big] = g^{-1}(X_i^\top\beta)$, for β.

Proof sketch. The goal is to show that the bootstrap variance estimate for $\hat{\beta}$ is asymptotically equivalent to the sandwich matrix estimator.

Step 1: Recall the asymptotic distribution of the M-estimator. In the real world, $\hat{\beta}$ solves

$$\frac{1}{N}\sum_{i=1}^{N}\psi(Y_i, X_i;\,\beta) = 0.$$

By M-estimation theory, we already know that $\sqrt{N}(\hat{\beta}-\beta_0) \xrightarrow{d} N(0,\,V)$ where $V = A(\beta_0)^{-1}B(\beta_0)\,A(\beta_0)^{-\top}$.

Step 2: The "true distribution" in the bootstrap world. The nonparametric bootstrap draws (X_i^*, Y_i^*) with replacement from the empirical distribution $\hat{F}_N$, then solves on each bootstrap sample:

$$\frac{1}{N}\sum_{i=1}^{N}\psi(Y_i^*, X_i^*;\,\beta) = 0 \quad\Rightarrow\quad \hat{\beta}^*.$$

The key observation is that in the bootstrap world, $\hat{F}_N$ plays the role of the "true distribution" and $\hat{\beta}$ plays the role of the "true parameter" β_0.

Step 3: Apply M-estimation theory in the bootstrap world. The asymptotic normality theorem for M-estimators holds for any data generating distribution satisfying regularity conditions. So in the bootstrap world, conditional on the original data:

$$\sqrt{N}(\hat{\beta}^* - \hat{\beta}) \xrightarrow{d^*} N(0,\,V^*)$$

where $V^* = A^*(\hat{\beta})^{-1}\,B^*(\hat{\beta})\,A^*(\hat{\beta})^{-\top}$. Here A^* and B^* are expectations under the empirical distribution $\hat{F}_N$:

$$A^*(\hat{\beta}) = E_{\hat{F}_N}\big[\dot{\psi}(Y, X;\,\hat{\beta})\big] = \frac{1}{N}\sum_{i=1}^{N}\dot{\psi}(Y_i, X_i;\,\hat{\beta}) = \hat{A}$$

good luck :)

$$B^*(\hat\beta) = E_{\hat F_N}\big[\psi(Y,X;\,\hat\beta)\,\psi(Y,X;\,\hat\beta)^\top\big] = \frac{1}{N}\sum_{i=1}^{N}\psi(Y_i,X_i;\,\hat\beta)\,\psi(Y_i,X_i;\,\hat\beta)^\top = \hat B.$$

Step 4: Recognize the sandwich estimator. Notice that $\hat A$ and $\hat B$ are exactly the plug-in estimators of the sandwich matrix. Therefore:

$$V^* = \hat A^{-1}\,\hat B\,\hat A^{-\top} = \hat V_{\text{sandwich}}.$$

Step 5: Consistency. By the LLN and $\hat\beta \xrightarrow{p} \beta_0$, under appropriate regularity conditions:

$$\hat A \xrightarrow{p} A(\beta_0), \qquad \hat B \xrightarrow{p} B(\beta_0)$$

$$\Rightarrow \quad V^* = \hat V_{\text{sandwich}} \xrightarrow{p} V.$$

Summary. The bootstrap variance is equivalent to the sandwich matrix because the bootstrap treats the empirical distribution as the "true distribution" and re-runs the M-estimation asymptotic expansion under it. The resulting A^* and B^* are naturally the sample versions of A and B. This is not a coincidence, but a direct manifestation of bootstrap consistency: the bootstrap simulates the sampling distribution of $\hat\beta$, and M-estimation theory tells us that the variance of this sampling distribution is precisely the sandwich form.

Link GLM with M-Estimators

Recall that generalized linear models estimated by maximum likelihood solve the score equation

$$s(\beta) = \sum_{i=1}^{N}\frac{\omega_i(Y_i-\mu_i)}{\phi\,g'(\mu_i)\,V(\mu_i)}\,X_i \stackrel{\text{set}}{=} 0,$$

where $\mu_i = g^{-1}(X_i^\top\beta)$. We derived this from the likelihood (say, Poisson, or binomial, or gamma, or whatever). **Question:** is this a reasonable set of estimating equations in general — i.e., if we did not know that this was associated to maximizing a likelihood, would it still make sense to use? Consider the population-level version:

$$\mathbb{E}\Big[\frac{\omega_i(Y_i-\mu_i)}{g'(\mu_i)\,V(\mu_i)}\,X_i\Big] = 0.$$

As long as (i) $\mathbb{E}\big[Y_i \mid X_i\big] = \mu_i$ where $g(\mu_i) = X_i^\top\beta$ and (ii) the errors $Y_i - \mu_i$ are independent (or uncorrelated) with the X_i's, then this equation must be true. The claim: **this set of estimating equations is valid under much more general conditions than those that it was derived under.**

Taking a step back, if we are assuming that the model $\mathbb{E}\big[Y_i \mid X_i = x\big] = g^{-1}(X_i^\top\beta_0)$ for some β_0, **but not assuming anything about the variance** or that the observations come from an exponential dispersion family, then a natural estimating equation for accomplishing this goal is

$$\mathbb{E}\big[A(X_i)(Y_i-\mu_i)\big] = \mathbf{0}$$

for an arbitrary choice of $A : \mathcal{X} \to \mathbb{R}^P$. As a special case, we can set $A(X_i) = \frac{\omega_i\,X_i}{g'(\mu_i)\,V(\mu_i)}$, and so we see that the MLE we derived for a generalized linear model actually corresponds to a very specific choice of M-estimator.

<u>Remark.</u> The two conditions for the estimating equation to hold at the population level: (i) $E[Y_i \mid X_i] = \mu_i$ and (ii) $Y_i - \mu_i$ is uncorrelated with X_i. In fact, condition (i) alone suffices, since $E[Y_i - \mu_i \mid X_i] = 0$ implies $\mathrm{Cov}(Y_i - \mu_i,\, h(X_i)) = 0$ for *any* function h by iterated expectations. Conversely, condition (ii) alone is *not* sufficient: uncorrelatedness with X_i does not guarantee uncorrelatedness with the nonlinear function $\frac{\omega_i}{g'(\mu_i)\,V(\mu_i)}\,X_i$ that appears in the estimating equation. The correct minimal assumption is therefore simply that the mean structure is correctly specified.

Quasi-Likelihood

A limitation of the M-estimators we have studied so far is that they essentially produce "Wald" methods and, as we have seen earlier, Wald methods are not very good.
Interestingly, it turns out that we can get a very convenient likelihood-adjacent framework by making only the **two moment assumptions**:

$$\mathbb{E}\big[Y_i \mid X_i = x_i\big] = \mu_i,$$
$$\mathrm{Var}(Y_i \mid X_i = x_i) = \frac{\phi}{\omega_i} V(\mu_i),$$

where $g(\mu_i) = x_i^\top \beta$ and $V(\mu_i)$ are specified by the user. This is a stronger assumption than what is required for the validity of an M-estimator, but weaker than what is needed when assuming that a generalized linear model is correctly specified.
The interesting thing about making these assumptions is that we are able to build a function that **behaves** like it was a likelihood. We define the quasi-density/mass function:

$$q(y \mid \phi, \mu) = \exp\Big\{\frac{1}{\phi}\int_y^\mu \frac{y-t}{V(t)}\,dt\Big\}.$$

Based on this, we can construct a data-level quasi-likelihood:

$$Q(\beta) = \prod_i q(Y_i \mid \phi/\omega_i,\, \mu_i) \qquad \text{where} \quad g(\mu_i) = X_i^\top \beta.$$

If you just pretend that $q(y \mid \phi, \mu)$ is a likelihood, even though it is not, everything is fine: Wilks' theorem still holds for quasi-likelihoods, so you still get valid likelihood ratio tests and score tests. The dispersion parameter can be estimated in the same way as before:

$$\hat{\phi} = \frac{1}{N-P}\sum_i \frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}.$$

Where does this idea come from? Starting from the observation that

$$s(\beta) = \sum_i \frac{\omega_i (Y_i - \mu_i)}{\phi\, V(\mu_i)\, g'(\mu_i)} X_i \overset{\text{set}}{=} 0$$

is the score equation for a GLM, one can ask the question: what likelihood would produce this as its score equation? The answer is exactly the quasi-likelihood. **It means the score function $s(\beta)$ of Quasi-likelihood is same as the expression formula of GLM's.**

Quasi-Poisson

The quasi-Poisson model sets $V(\mu) = \mu$ (the same variance function as the Poisson), but without the restriction that $\phi = 1$.

Quasi-Binomial

The quasi-Binomial model sets $V(\mu) = \mu(1-\mu)$ (the same variance function as the binomial), again without the restriction that $\phi = 1$.

Why Quasi-Likelihood?

From the previous batches of notes, we have seen that we already have models for count and binomial-type data that allow for overdispersion, which is also the main selling point of quasi-likelihood methods. A couple of benefits of quasi-likelihood methods are that:

1. They are generally easier to fit, and more reliable, than (say) negative binomial or beta-binomial regression models. The latter models run into computational issues a bit more easily.

2. As default models for the variance, many folks prefer to have the variance in count data to increase *linearly* in the mean, i.e., $\text{Var}(Y_i \mid X_i) = \phi\,\mu_i$; they just want to have $\phi \neq 1$. By contrast, the negative binomial model takes $\text{Var}(Y_i \mid X_i = x) = \mu_i + \mu_i^2/k$, which is quadratic in the mean. This increases pretty quickly, and this level of overdispersion is often unrealistic.

3. Quasi-likelihood works very well as a post-hoc fix to the Poisson or logistic regression models. The **point estimates** of the quasi-Poisson and quasi-Binomial are **exactly the same** as the point estimates of the Poisson and the binomial. The only difference is in inference: instead of an inverse Fisher information of $(X^\top DX)^{-1}$ for the Poisson/binomial, we now have $\hat{\phi}(X^\top DX)^{-1}$.

Empirical Likelihood

The selling point of quasi-likelihood, relative to just treating $\hat{\beta}$ as an M-estimator, is that we can easily get something resembling likelihood-based inference from the quasi-likelihood. This is not possible to get just from using robust standard errors, where we are mostly limited to something resembling Wald-type methods. A natural follow-up question is then: "is it possible to get something that behaves like a likelihood function even if we do not make any assumptions about the variance?" It turns out that the answer is "yes, by using the *empirical likelihood*."

Motivation: Nonparametric MLE. Suppose $Z_1, \dots, Z_N \overset{\text{iid}}{\sim} F_0$. Our goal is to estimate F_0 nonparametrically. A natural estimator is the empirical distribution $\hat{F}_N = \frac{1}{N}\sum_{i=1}^N \delta_{Z_i}$. We can motivate this as an MLE in two steps.

Step 1: Reduce the class of distributions. We restrict attention to distributions supported on the observed data, i.e., $F = \sum_i p_i\,\delta_{Z_i}$, where $p_i \geq 0$ and $\sum_i p_i = 1$. Under such a distribution, the likelihood of the observed data is

$$L(F) = P_F(Z_1 = z_1) \cdot P_F(Z_2 = z_2) \cdots P_F(Z_N = z_N) = \prod_{i=1}^N p_i.$$

Step 2: Optimize over $p_1, \dots, p_N$. We maximize the log-likelihood $\ell(F) = \sum_{i=1}^N \log p_i$ subject to $\sum_i p_i = 1$, $p_i \geq 0$. Using a Lagrange multiplier for the sum constraint:

$$\ell^*(F;\, \lambda) = \sum_{i=1}^N \log p_i - \lambda\big(\sum_i p_i - 1\big).$$

Differentiating with respect to p_i and setting to zero gives $\frac{1}{p_i} - \lambda = 0$, so $p_i = 1/\lambda$. Substituting into $\sum p_i = 1$ yields $N/\lambda = 1$, hence $\lambda = N$ and $p_i = 1/N$ for all i. This recovers the empirical distribution $\hat{F}_N$.

Adding an Estimating Equation Constraint. Now suppose we believe that the true β_0 satisfies $E_{F_0}[\psi(Y, X;\, \beta)] = 0$ for some estimating equation ψ. Under a discrete distribution $F = \sum_i p_i\,\delta_{(X_i, Y_i)}$, the expectation is simply a weighted sum:

$$E_F\big[\psi(Y, X;\, \beta)\big] = \sum_i p_i\,\psi_i(\beta).$$

We therefore add this as an additional constraint: for a given β, maximize $\prod p_i$ subject to $\sum_i p_i = 1$, $p_i \geq 0$, and $\sum_i p_i \, \psi_i(\beta) = 0$. Profiling out the p_i's gives the profile empirical likelihood.
Definition. The *profile empirical likelihood* of β is given by

$$L_{\mathrm{EL}}(\beta) = \max\Big\{\prod_{i=1}^{N} p_i : \sum_i p_i \frac{\omega_i (Y_i - \mu_i)\, X_i}{\phi\, V(\mu_i)\, g'(\mu_i)} = 0,\ p_i \geq 0,\ \sum_i p_i = 1\Big\}.$$

Under standard regularity conditions, a variant of Wilks' theorem holds:

$$-2\big\{\ell_{\mathrm{EL}}(\beta_0) - \ell_{\mathrm{EL}}(\hat{\beta}_{\mathrm{EL}})\big\} \xrightarrow{d} \chi^2_P$$

where $\ell_{\mathrm{EL}} = \log L_{\mathrm{EL}}$, β_0 is the true parameter value, and $\hat{\beta}_{\mathrm{EL}}$ is the maximum empirical likelihood estimator. This result allows for likelihood-based inference (such as confidence regions and hypothesis tests) without requiring parametric assumptions about the distribution of the data or specification of the variance function.

Summary: M-Estimation vs. Quasi-Likelihood vs. Empirical Likelihood

M-Estimation. Requires only a correct mean specification: $\mathbb{E}\big[Y_i \mid X_i\big] = g^{-1}(X_i^\top \beta)$. No assumptions are needed on the variance or the distributional family. The estimator $\hat{\beta}$ is consistent and asymptotically normal with sandwich variance $V = A(\beta_0)^{-1} B(\beta_0)\, A(\beta_0)^{-\top}$. The drawback is that inference is limited to **Wald-type methods** — point estimates plus sandwich (or bootstrap) standard errors. There is no likelihood function, so likelihood ratio tests and analysis of deviance are not available.
Quasi-Likelihood. Requires both a correct mean specification $\mathbb{E}\big[Y_i \mid X_i\big] = g^{-1}(X_i^\top \beta)$ **and** a correct variance specification $\mathrm{Var}(Y_i \mid X_i) = \frac{\phi}{\omega_i} V(\mu_i)$. This is stronger than what M-estimation requires, but weaker than assuming a full exponential dispersion family. The payoff is that we can construct a quasi-likelihood function $Q(\beta)$ that **behaves like a real likelihood**: Wilks' theorem holds, so we get likelihood ratio tests, analysis of deviance (via F-tests), and score tests, not just Wald inference.
Empirical Likelihood. Requires only a correct mean specification — the same assumption as M-estimation. No variance assumption is needed. Despite this, the profile empirical likelihood $L_{\mathrm{EL}}(\beta)$ **also behaves like a real likelihood**: a variant of Wilks' theorem holds, giving us likelihood ratio–type tests and confidence regions. In other words, empirical likelihood achieves the inferential benefits of quasi-likelihood (going beyond Wald) while requiring only the minimal assumptions of M-estimation.

	M-Estimation	Quasi-Likelihood	Empirical Likelihood
Mean assumption	✓	✓	✓
Variance assumption		✓	
Distributional family			
Wald inference	✓	✓	✓
LRT / Deviance		✓	✓

good luck :)

8 Survival Analysis

Survival analysis concerns modeling the time until a particular event occurs. This is common in clinical trials (time-to-death, time-to-adverse-event), reliability analysis (time-to-failure), and logistics (time-to-shipment).

Basic Setup and Key Quantities

Notation. Let T_i denote the survival time, C_i the censoring time, $Y_i = \min(T_i, C_i)$ the observed event time, $\delta_i = \mathbf{1}(T_i \le C_i)$ the censoring indicator (1 if we observe the true event, 0 if censored), and X_i the covariates. The observed data for each individual is the triple (Y_i, δ_i, X_i). WLOG, assume that $Y_1 < Y_2 < \cdots < Y_N$.
Survival Function. The primary target is the *survival function*:

$$S(t \mid x) = \Pr(T_i > t \mid X_i = x) = 1 - \Pr(T_i \le t \mid X_i = x).$$

Hazard Function. It is often convenient to instead work with the *hazard function*:

$$h(t \mid x) = \frac{d}{dt}\big[-\log S(t \mid x)\big] = \frac{f(t \mid x)}{S(t \mid x)} = \lim_{\Delta \to 0} \frac{\Pr(T_i \le t + \Delta \mid T_i > t)}{\Delta}.$$

The hazard is such that, for some tiny time window Δ, the probability of dying between t and $t+\Delta$ given survival past time t is approximately $h(t \mid x) \times \Delta$. Working with the hazard is convenient because it naturally handles right-censored data and is a natural place to incorporate covariate dependence.
<u>**Remark.**</u> The three expressions for the hazard are all equivalent. Starting from $S(t) = \Pr(T > t)$, we have $f(t) = -S'(t)$, so $\frac{f(t)}{S(t)} = \frac{-S'(t)}{S(t)} = -\frac{d}{dt}\log S(t) = \frac{d}{dt}[-\log S(t)]$. The limit definition follows from $\Pr(T \le t + \Delta \mid T > t) = \frac{S(t) - S(t+\Delta)}{S(t)} \approx \frac{f(t)\Delta}{S(t)}$.
Cumulative Hazard. The *cumulative hazard function* is $H(t \mid x) = \int_0^t h(z \mid x)\, dz = -\log S(t \mid x)$, so that $S(t \mid x) = \exp\big\{-H(t \mid x)\big\}$.

<u>**Remark.**</u> Although both $F(t) = \Pr(T \le t) = 1 - S(t)$ and $H(t) = \int_0^t h(z)\, dz = -\log S(t)$ are cumulative quantities that increase with t, they are fundamentally different:

- $F(t)$ is a **probability** (the probability that the event has occurred by time t), so $F(t) \in [0, 1]$.
- $H(t)$ is the **accumulated risk** up to time t, obtained by integrating the instantaneous hazard. It is *not* a probability, and $H(t) \in [0, \infty)$.

They are linked via the transformation $F(t) = 1 - e^{-H(t)}$. No matter how large $H(t)$ grows, the exponential map compresses it back into $[0, 1]$. Intuitively, $H(t)$ is like the total "danger" you have accumulated walking along a road; $F(t)$ is the probability that this accumulated danger has actually caused the event to occur.

Right-Censoring

In most survival settings, the survival time is *right-censored*: rather than observing T_i directly, we sometimes only know that $T_i > C_i$ for some censoring time C_i (e.g., the study ends before the event occurs). We never observe both T_i and C_i; we only observe whichever occurs first.
The Survival Likelihood. Suppose $\big\{S_\theta : \theta \in \Theta\big\}$ is a parametric model for the survival function with associated hazards $h_\theta(t \mid x)$, and assume that $C_i \perp\!\!\!\perp (T_i, X_i)$. Then the likelihood is

$$L(\theta) = \prod_{i=1}^{N} h_\theta(Y_i \mid X_i)^{\delta_i}\, S_\theta(Y_i \mid X_i) = \prod_{i=1}^{N} h_\theta(Y_i \mid X_i)^{\delta_i} \exp\big\{-\int_0^{Y_i} h_\theta(y \mid X_i)\, dy\big\}.$$

The key intuition: uncensored observations ($\delta_i = 1$) contribute the density $f = h \cdot S$ at Y_i, while censored observations ($\delta_i = 0$) contribute only the survival probability $S(Y_i)$, reflecting the information that $T_i > Y_i$.

Kaplan-Meier Estimator

Before discussing regression, we consider nonparametric estimation of $S(t)$ under censoring. Suppose our data consists of $(Y_1, \delta_1), \ldots, (Y_N, \delta_N)$ with no covariates.

Derivation of the Kaplan-Meier Estimator.

(a) Product decomposition of $S(t)$. For any times $s < t$, the survival function can be written as

$$S(t) = S(s) \times \Pr(T > t \mid T > s).$$

Proof. By the definition of conditional probability, $\Pr(T > t \mid T > s) = \frac{\Pr(T>t, T>s)}{\Pr(T>s)}$. Since $t > s$, the event $\{T > t\}$ implies $\{T > s\}$, so $\Pr(T > t, T > s) = \Pr(T > t)$. Hence $\Pr(T > t \mid T > s) = \frac{S(t)}{S(s)}$, from which $S(t) = S(s) \times \Pr(T > t \mid T > s)$.

Applying this recursively, for any sequence of times $0 = t_0 < t_1 < \cdots < t_k = t$, we can write

$$S(t) = \prod_{j=1}^{k} \Pr(T > t_j \mid T > t_{j-1}).$$

(b) No censoring case. For a dataset where there is no censoring (i.e., $\delta_i \equiv 1$), taking t_j to be the observed event times and estimating each conditional probability with the empirical frequencies yields

$$\hat{S}(t) = \frac{\text{number of observations} > t}{N}.$$

That is, we recover the empirical survival function.

Proof. Empirically, we can write $S(t)$ as:

$$\hat{S}(t) = \prod_{j=1}^{k} \frac{\text{observations} > t_j}{\text{observations} > t_{j-1}} = \prod_{j=1}^{k} \frac{\text{observations} > t_{j-1} - \text{deaths at } t_j}{\text{observations} > t_{j-1}} = \frac{\text{observations} > t_j}{\text{observations} > t_0} = \frac{\text{observations} > t}{N}.$$

Note that this equality only holds when the observations survived past t_{j-1} is still observable (uncensored) and counted in t_j.

(c) Estimation under independent censoring. When independent censoring is present (i.e., C_i and T_i are independent), $\Pr(T > t_j \mid T > t_{j-1})$ can be estimated with $(n_j - d_j)/n_j$, where n_j is the number of individuals at risk just before t_j and d_j is the number of deaths at t_j. Why is this sensible?

Proof. Recall that

- n_j is the number of individuals who were at risk of dying in the study at time $t_{(j)}$ (those who have neither died nor censored). For a subject, it corresponds to the event $\{T_i > t_{(j-1)} \cap C_i \geq t_{(j)}\}$.
- d_j is the number of deaths occurred at time $t_{(j)}$.

Our goal is to prove why the following equation works: $\widehat{\Pr}(T > t_{(j)} \mid T > t_{(j-1)}) \stackrel{?}{=} \frac{\widehat{\Pr}(T>t_{(j)}, C \geq t_{(j)})}{\widehat{\Pr}(T>t_{(j-1)}, C \geq t_{(j)})} = \frac{(n_j - d_j)/N}{n_j/N} = \frac{n_j - d_j}{n_j}$.

Now, invoking the independence censoring assumption ($C_i \perp\!\!\!\perp T_i$) and utilizing distinct failure times $t_{(1)} \dots t_{(k)}$, we can write

$$\begin{aligned}
\Pr(T > t_{(j)} \mid T > t_{(j-1)}) &= \Pr(T > t_{(j)} \mid T > t_{(j-1)},\, C \geq t_{(j)}) \\
&= \frac{\Pr(T > t_{(j)},\, T > t_{(j-1)},\, C \geq t_{(j)})}{\Pr(T > t_{(j-1)},\, C \geq t_{(j)})} \\
&= \frac{\Pr(T > t_{(j)},\, C \geq t_{(j)})}{\Pr(T > t_{(j-1)},\, C \geq t_{(j)})}
\end{aligned}$$

This estimator is only sensible under the assumption that $C_i \perp\!\!\!\perp T_i$. Because the number of patients who survived past $t_{(j-1)}$ is not necessarily equivalent to those observed at $t_{(j)}$ due to censoring.
(d) Combining (a) and (c) yields the KM estimator. Substituting the estimates from part (c) into the product formula from part (a), with t_j taken to be the distinct observed *failure* times $t_{(1)} < t_{(2)} < \cdots < t_{(K)}$, we obtain the Kaplan-Meier estimator:

$$\hat{S}(t) = \prod_{j:\, t_{(j)} \leq t} \big(1 - \frac{d_j}{n_j}\big).$$

R Implementation. The `survival` package provides `Surv()` and `survfit()`. Censored observations are printed with a + symbol. The `summary(km_fit, times = c(...))` function gives survival estimates with pointwise confidence intervals at specified timepoints. To plot separate KM curves by group, use `survfit(Surv(time, event) ~ group)`.

Logrank Test

To test whether K groups have the same survival function, i.e., $H_0 : S_1(t) = \cdots = S_K(t)$ for all t versus H_1: at least one $S_k(t)$ differs, we use the *logrank test*. The test statistic Q has an asymptotic χ^2_{K-1} distribution under H_0. In R: `survdiff(Surv(time, status) ~ group, data = ...)`.
Remark. The logrank test statistic is the same as the score test statistic for the Cox proportional hazards model with the group variable as the covariate. So for two groups, the logrank test and the score test of a Cox model with a single binary covariate are equivalent.

Cox's Proportional Hazards Model

The standard way to incorporate covariates into survival analysis (although not the only way, just the most popular) is the Cox proportional hazards model. In its most general form:

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline hazard}} \times \underbrace{\Psi(x^\top \beta)}_{\text{link function}}.$$

It is called *proportional hazards* because the ratio of hazards for two individuals does not depend on time:

$$\frac{h(t \mid x)}{h(t \mid x')} = \frac{\Psi(x^\top \beta)}{\Psi((x')^\top \beta)}.$$

The standard choice is $\Psi(\eta) = e^\eta$, which gives

$$h(t \mid x) = h_0(t)\, \exp\!\big(x^\top \beta\big).$$

good luck :)

Interpreting Coefficients. Taking $\Psi(\eta) = e^{\eta}$, the hazard ratio simplifies to

$$h(t \mid x') = h(t \mid x) \times \exp\big((x' - x)^\top \beta\big).$$

Hence e^{β_j} is the *hazard ratio* associated with a one-unit increase in x_j, holding all other covariates fixed. If $e^{\beta_j} > 1$, increasing x_j increases the hazard (worse prognosis); if $e^{\beta_j} < 1$, it decreases the hazard (better prognosis).
Remark. We do not include an intercept in X_i because any intercept β_0 would be absorbed into $h_0(t)$: writing $h_0(t)\, e^{\beta_0 + x^\top \beta} = \big[h_0(t)\, e^{\beta_0}\big] e^{x^\top \beta}$, the bracket is simply a new baseline hazard.
Survival Function Under the Cox Model. Under proportional hazards, $S(t \mid x) = S_0(t)^{\exp(x^\top \beta)}$ where $S_0(t) = \exp\big(-\int_0^t h_0(z)\, dz\big)$ is the baseline survival function. Equivalently, this gives a log-log link:

$$\log\big\{-\log S(t \mid x)\big\} = \log\big\{-\log S_0(t)\big\} + x^\top \beta.$$

Visual Check of PH Assumption. For a binary or categorical covariate x, the proportional hazards assumption implies that $\log\{-\log \hat{S}(t \mid x)\}$ versus $\log t$ should produce **parallel curves** across groups. Crossing or diverging curves suggest that the PH assumption may be violated.

Likelihood Under Proportional Hazards

Plugging $h(t \mid x) = h_0(t)\, e^{x^\top \beta}$ into the survival likelihood gives

$$L(h_0, \beta) = \prod_{i=1}^{N} \big\{h_0(Y_i)\, \exp\big(X_i^\top \beta\big)\big\}^{\delta_i} \times \exp\Big\{-\exp\big(X_i^\top \beta\big) \int_0^{Y_i} h_0(t)\, dt\Big\}.$$

This is useful for parametric models. **But** there is an inconvenience: in order to estimate β, we have to simultaneously estimate $h_0(t)$, which is an infinite-dimensional nuisance parameter. Is there a way to estimate β without having to estimate $h_0(t)$?

Cox's Partial Likelihood: The Trick

Cox's key insight was that β can be estimated *without* directly estimating $h_0(t)$, by conditioning on the observed failure times and asking: "given that we know a death occurs at time $t_{(k)}$, what is the probability that the specific individual who died is individual i?"
Derivation. Order the observed failure times $t_{(1)} < t_{(2)} < \cdots < t_{(K)}$ (assuming no ties). At time $t_{(k)}$, exactly one individual dies. For a given individual i, there are three possibilities:

1. If individual i is censored before time $t_{(k)}$, then it is impossible that this individual died at $t_{(k)}$ — if they had, we would not have observed a censoring.

2. If individual i has already died prior to $t_{(k)}$, then it is also impossible.

3. If individual i is both alive and uncensored at time $t_{(k)}$, then their infinitesimal probability of dying at $t_{(k)}$ is $h_0(t_{(k)})\, e^{X_i^\top \beta}\, dt_k$.

Given that exactly one individual is known to die at time $t_{(k)}$, the probability that it is specifically individual i is obtained by normalizing over all at-risk individuals:

$$\frac{h_0(t_{(k)})\, \exp\big(X_i^\top \beta\big)\, dt_k}{\sum_{i' \in \mathcal{R}(t_{(k)})} h_0(t_{(k)})\, \exp\big(X_{i'}^\top \beta\big)\, dt_k} = \frac{\exp\big(X_i^\top \beta\big)}{\sum_{i' \in \mathcal{R}(t_{(k)})} \exp\big(X_{i'}^\top \beta\big)}$$

where the *risk set* $\mathcal{R}(t_{(k)}) = \{i : Y_i \geq t_{(k)}\}$ is the collection of all individuals who are alive and uncensored at time $t_{(k)}$. The baseline hazard $h_0(t_{(k)})$ and the dt_k cancel completely! Based on this, the *partial likelihood* is the probability that we saw the particular sequence of failures that we did:

$$\mathrm{PL}(\beta) = \prod_{k=1}^{K} \frac{\exp\bigl(X_{(k)}^\top \beta\bigr)}{\sum_{i \in \mathcal{R}(t_{(k)})} \exp\bigl(X_i^\top \beta\bigr)}$$

where $X_{(k)}$ is the covariate vector of the individual who actually died at $t_{(k)}$.
Log-Partial Likelihood. Equivalently, summing over the uncensored observations:

$$\ell_{\mathrm{Cox}}(\beta) = \sum_{i:\, \delta_i = 1} \Bigl[X_i^\top \beta - \log \sum_{i' :\, Y_{i'} \geq Y_i} \exp\bigl(X_{i'}^\top \beta\bigr)\Bigr].$$

Remark. The partial likelihood $\ell_{\mathrm{Cox}}(\beta)$ behaves like a genuine log-likelihood: it can be used for likelihood ratio tests, Wald tests, and score tests, and the inverse Fisher information gives the asymptotic covariance of $\hat{\beta}$. This is the foundation of semiparametric inference in survival analysis — we get valid inference for β without ever specifying a parametric form for $h_0(t)$.

Estimating the Baseline Survival

After estimating β via partial likelihood, if we want the survival function we go through the route of estimating the *cumulative* baseline hazard $H_0(t) = \int_0^t h_0(z)\, dz$ rather than $h_0(t)$ itself (estimating $h_0(t)$ directly via nonparametric MLE places infinite mass at the observed failure times, analogous to estimating a density via MLE with no restrictions). The *Breslow estimator* is

$$\hat{H}_0(t) = \sum_{j:\, t_{(j)} \leq t} \frac{d_j}{\sum_{l \in \mathcal{R}(t_{(j)})} \exp\bigl(x_l^\top \hat{\beta}\bigr)}$$

where d_j is the number of events at time $t_{(j)}$ and $\mathcal{R}(t_{(j)}) = \{\ell : Y_\ell \geq t_{(j)}\}$. The estimated baseline survival function is $\hat{S}_0(t) = \exp\{-\hat{H}_0(t)\}$, and the estimated survival function for an individual with covariates x is

$$\hat{S}(t \mid x) = \exp\bigl\{-\hat{H}_0(t)\, \exp\bigl(x^\top \hat{\beta}\bigr)\bigr\}.$$

Concordance (C-Index)

The usual goodness-of-fit measure for survival models is the *concordance* (C-index), which measures how well the model ranks individuals by risk.
Intuition. Suppose we have two individuals i and i' with risk scores $\eta_i = X_i^\top \hat{\beta}$ and $\eta_{i'} = X_{i'}^\top \hat{\beta}$. If $\eta_i > \eta_{i'}$ (individual i is at higher risk), then we should expect $T_i < T_{i'}$ (individual i dies first). We call such a pair **concordant** if the model's ranking agrees with the observed ordering. The concordance is essentially the proportion of pairs where the model "guesses correctly."
Definition. A pair (i, i') with $Y_i < Y_{i'}$ is *comparable* if $\delta_i = 1$ (we know for certain that subject i died before i'). A comparable pair is *concordant* if $X_i^\top \hat{\beta} > X_{i'}^\top \hat{\beta}$. The C-index is

$$C = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}} = \frac{\sum_{i,i'} \delta_i\, \mathbf{1}(Y_i < Y_{i'})\, \mathbf{1}(X_i^\top \hat{\beta} > X_{i'}^\top \hat{\beta})}{\sum_{i,i'} \delta_i\, \mathbf{1}(Y_i < Y_{i'})}.$$

Formally, C estimates $\Pr(X_1^\top \beta > X_2^\top \beta \mid T_1 < T_2, \delta_1 = 1)$. A value of $C = 0.5$ indicates no predictive power (random guessing), while $C = 1$ indicates perfect discrimination. This metric is conceptually similar to AUC in binary classification, adapted for censored data.

Fitting Cox Regression in R

The `coxph()` function fits Cox proportional hazards models:

```
cox_model <- coxph(Surv(time, status) ~ x1 + x2 + ..., data = ...).
```

Output Interpretation. The `summary()` output reports: (1) estimated coefficients $\hat{\beta}$ (`coef` column); (2) hazard ratios $e^{\hat{\beta}}$ (`exp(coef)` column); (3) Wald standard errors and P-values; (4) the concordance; (5) three global tests — likelihood ratio, Wald, and score (logrank) — which are all asymptotically equivalent under H_0 but may differ in finite samples. As with GLMs, the likelihood ratio test is generally preferred.

Variable Selection and Nested Models. The `drop1(cox_model, test = "Chisq")` function tests each variable's significance via partial-likelihood ratio tests. The `anova()` function can compare nested Cox models: for example, to test whether an interaction `trt * celltype` is significant, fit both the full model (`karno + age + trt * celltype`) and the reduced model (`karno + age + celltype`), then call `anova(full, reduced)` to obtain a χ^2 test.

Individualized Survival Curves. After fitting a Cox model, `survfit(cox_model, newdata = ...)` generates predicted survival curves for specific covariate profiles. These are particularly useful in clinical settings for giving patient-specific prognoses and informing treatment decisions.

9 Output Interpretation

good luck :)